

Jolly Helmet

Make your Helmet Happy Again!

Clean it. Dry it.
Wear it happy.

Coin-operated helmet cleaning and drying stations

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DEVELOPERS AND DESIGNERS

USER MANUAL

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Customer guide

Welcome to Jolly Helmet!

A self-service station that cleans and dries your motorcycle helmet while you wait.

Meet the machines

There are three Jolly Helmet stations. Each one has a touch screen, a coin slot and a door. Pick the one that fits what your helmet needs.

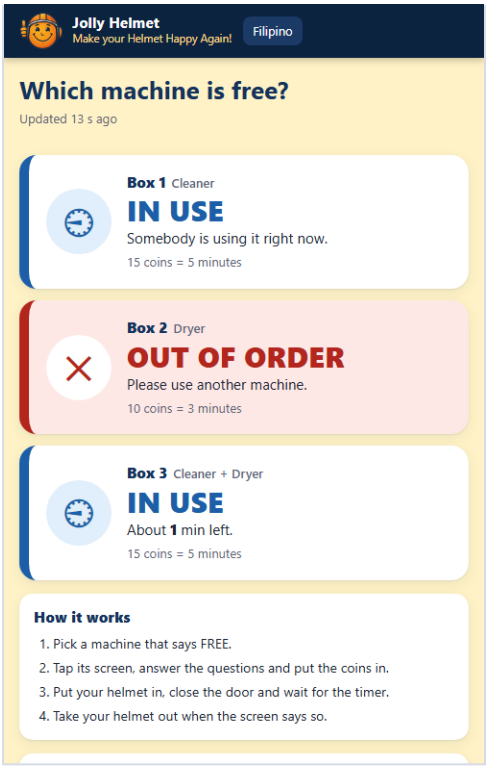


Station	What it does
Box 1 Cleaner	Removes odor. UV light plus a fine mist of alcohol, scented liquid, or both.
Box 2 Dryer	Dries a wet helmet with warm air, then cools it down before you take it out.
Box 3 Cleaner + Dryer	One station, two modes. Choose CLEANING or DRYING on the first screen.

Tip / Paalala: Pwede mong gamitin ang Box 1 at Box 2 nang magkasunod: linis muna, tapos tuyo.

Which machine is free?

Look for the **QR sticker** on the machines. Scan it with your phone camera. A page opens that shows Box 1, Box 2 and Box 3 as **FREE**, **IN USE** (with about how many minutes are left) or **OUT OF ORDER**. You do not need an app or a password. The button at the top changes the language between English and Filipino.



The customer status page: which machine is free (phone)

Tip / Paalala: I-scan ang QR sticker para makita kung aling makina ang libre. Pindutin ang "Filipino" sa itaas ng page.

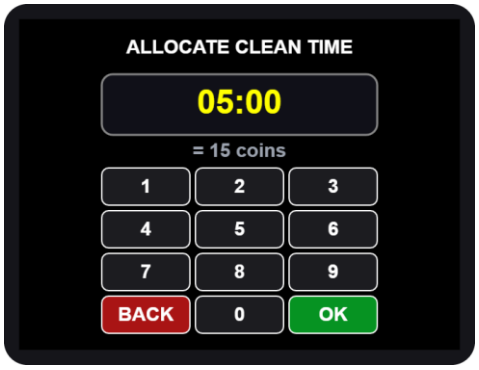
Before you start

You need: your helmet and coins. The machine counts coins by value: a one-peso coin counts as 1.

How the touch screen works



Rating grid: tap a box 1-10, then CONFIRM



Number pad: type the minutes and seconds, then OK

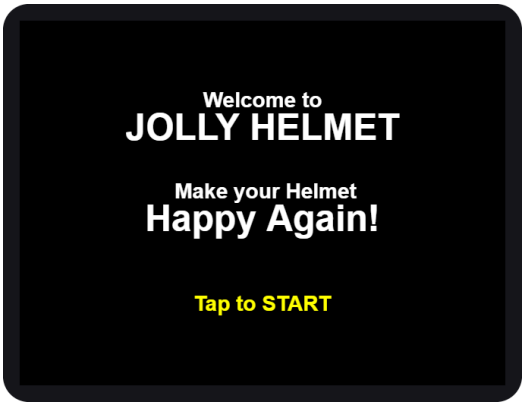
- **Big words** (yellow) tell you what to do next: OPEN, PLACE, CLOSE, TAKE.
- **Buttons** are the dark boxes with white words. Tap them once.
- The screen blinks briefly now and then. This is normal, it refreshes itself.

Never put your hand in while the machine runs. If you open the door, the machine pauses by itself.

Box 1 - Cleaner



1. Tap anywhere to START



2. Welcome

1. **Wake the machine.** Tap anywhere on the screen. For a moment it says "Please Wait" while it checks its alcohol and scented liquid.
2. **Tap to START** on the welcome screen.

> If the screen says **OUT OF ORDER**, the station needs refilling or service. Please tell the staff.

Rate it and choose the mist



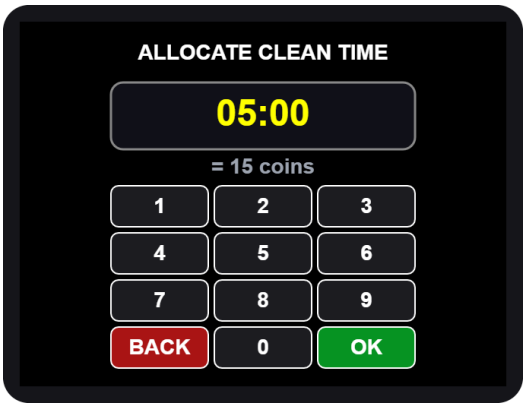
3. ODOR O' METER



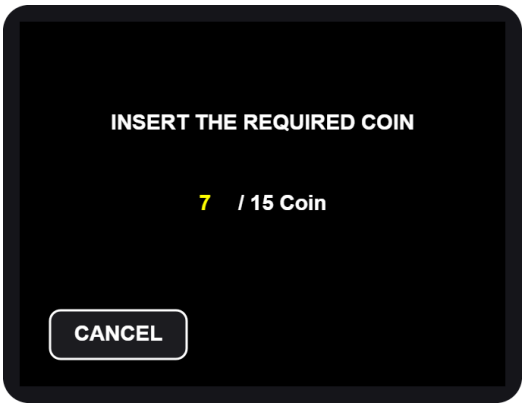
4. CLEANING MODE

3. **Rate the smell.** Tap 1 (MINT, fresh) to 10 (STINKY), then **CONFIRM**. The worse the smell, the more time we suggest.
4. **Choose the mist. BALANCED** uses both liquids, **MORE ALCOHOL** is stronger against germs, **MORE SCENTED** leaves a fresher scent.

Pick the time and pay



5. ALLOCATE CLEAN TIME



6. INSERT THE REQUIRED COIN

5. **Allocate time.** The pad shows the time we suggest for your rating. Type a different time if you like, then tap **OK**. The shortest time and the longest time are shown if you go outside them.
6. **Insert the coins.** The screen counts up: coins inserted / coins required. When it is enough, the machine moves on by itself.

Extra coins beyond the price give no extra time and are not refunded. Tap **CANCEL** before paying if you change your mind.

Load your helmet



7. OPEN



8. PLACE

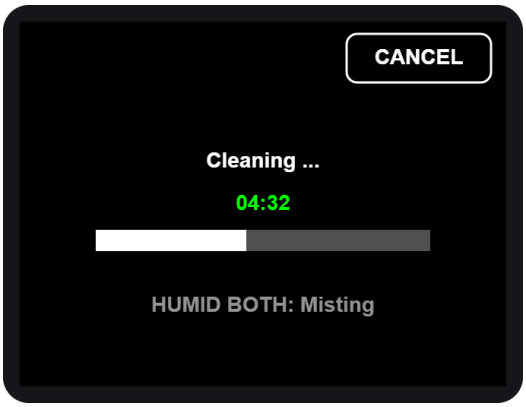


9. CLOSE

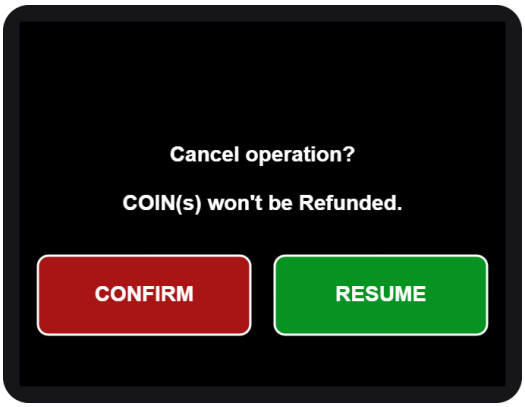
7. **OPEN** the door when the screen asks.
8. **PLACE** your helmet inside, opening facing the front, then step back.
9. **CLOSE** the door. The machine checks that the door is shut and the helmet is inside, then starts.

If you see **HELMET NO DETECTED**, the machine did not see your helmet. Open the door and place it again.

While it cleans



10. Cleaning... time left



CANCEL asks first

- The big green number is the **time left**. The bar fills as it goes.
- The small grey line shows what is on: the mist, the UV light.
- To stop early tap **CANCEL**, then **CONFIRM**. **Coins are not refunded. RESUME** keeps cleaning.

If the door is opened during a run, everything pauses (SAFETY PAUSE). Close the door and it continues where it stopped.

All done



11. TAKE your helmet



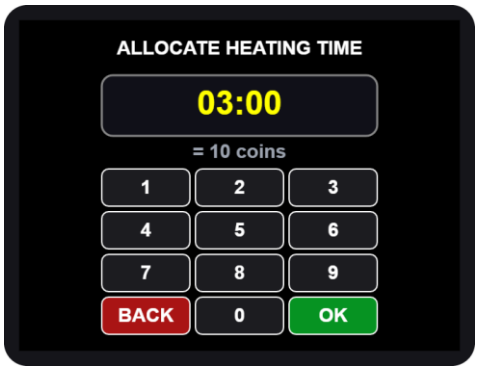
13. FREE TIME?

11. **OPEN** the door and **TAKE** your helmet. Close the door again.
12. **Rate it again** on the same 1-10 grid, then CONFIRM.
13. **Free retry.** Not happy yet? The machine may offer a short **FREE TIME** (2 minutes). Tap **YES** to run once more, or **NO**. It is offered once per visit.
14. The screen thanks you and returns to the start.

Box 2 - Dryer



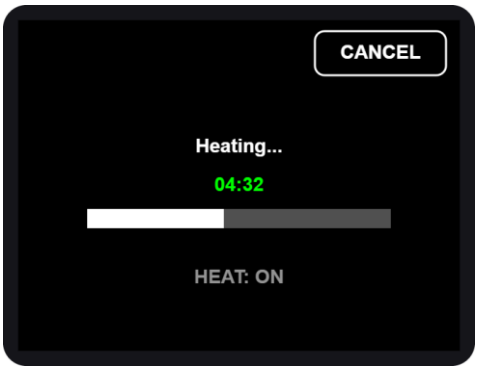
DRY O' METER



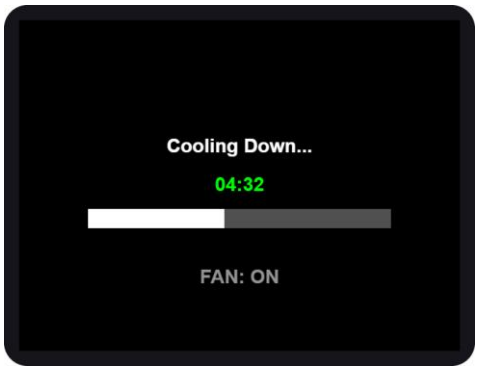
ALLOCATE HEATING TIME

1. **Tap anywhere**, then **Tap to START** (same as Box 1).
2. **Rate how wet it is:** 1 = DRY ... 10 = WET, then CONFIRM.
3. **Allocate heating time** on the number pad, then OK.
4. **Insert the coins** shown on the screen.
5. **OPEN, PLACE, CLOSE** the door, same as Box 1.

Heating and cooling



6. Heating...



7. Cooling Down...

6. **Heating.** Warm air dries the helmet and UV light freshens it. You can CANCEL, but coins are not refunded.
7. **Cooling down.** A fan runs alone so the helmet is not hot to wear. The door stays locked until it ends.
8. **OPEN** the door, **TAKE** your helmet, **CLOSE** the door.
9. **Rate it again,** then use the **free retry** if you want one.

If you cancel while heating, the machine still cools the helmet down first, for your safety.

Box 3 - Cleaner + Dryer



SELECT MODE



CLEANING unavailable

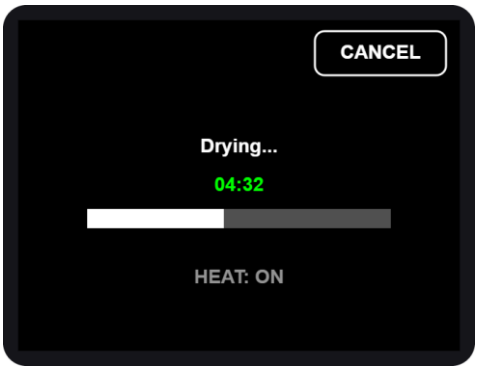
1. **Tap anywhere**, then **Tap to START**.
2. **SELECT MODE.** Tap **CLEANING** or **DRYING**.

If CLEANING is grey and says **UNAVAILABLE**, the cleaning liquids are being refilled or are low. **DRYING still works.**

CLEANING mode

Same steps as Box 1: rate the odor, choose the mist, pick the time, pay, load the helmet. At the end you rate **two things**: how dry it is, then how it smells. The heater runs only in the last minute, together with the UV light.

DRYING mode



Drying...



Cooling Down...

1. **Rate how wet** the helmet is (DRY O' METER), then CONFIRM.
2. **Allocate drying time** on the number pad.
3. **Pay, then OPEN, PLACE, CLOSE.**
4. **Drying** uses warm air and UV light. After it, the machine **cools down** for about a minute with nothing running.

5. **OPEN, TAKE, CLOSE,** rate again and use the free retry if offered. A free retry after cleaning is another cleaning; after drying it is another drying.

Prices and time

Box 1 and Box 3 (cleaning): 15 coins = 5 minutes.

Rating	Suggested time	Coins
1 - 5	5 minutes	15
6 - 7	10 minutes	30
8 - 9	15 minutes	45
10	20 minutes	60

Longest session: 30 minutes (90 coins).

Box 2 (drying): 10 coins = 3 minutes.

Rating	Suggested time	Coins (about)
1 - 3	3 minutes	10
4 - 6	5 minutes	17
7 - 8	8 minutes	27
9 - 10	10 minutes	34

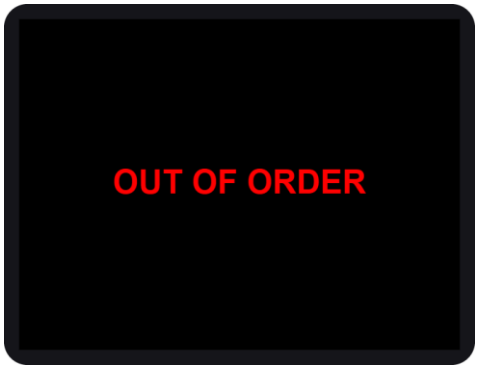
These are the starting prices. The admin can change them, so always check the **INSERT THE REQUIRED COIN** screen for the exact amount.

Special screens and safety

Special screens



Door opened during a run



Station needs service

You see	What it means / what to do
SAFETY PAUSE	The door was opened. Close it and the run continues.
HELMET NO DETECTED	No helmet seen. Open the door and place it again.
OUT OF ORDER	The station cannot work now. Use another station and tell the staff.
CLEANING UNAVAILABLE	Cleaning liquids are low. Use DRYING or come back later.
MAINTENANCE	A technician is working on the station.

Safety first

- **Helmets only.** Do not put bags, gloves, liquids or anything else inside.
- **Keep your hands out** while the machine runs. Hot air and UV light are used.
- **UV light** is strong. Never look into the chamber while it is on. The door must be closed.
- **Do not force the door.** It is locked on purpose while the machine runs. To stop, use **CANCEL** on the screen. If the screen does not answer, use the safety slide (see "If the door will not open").
- **Remove your helmet's accessories** (camera, speaker, visor tint film) if they are not heat-safe.
- **Children** should use the machine with an adult.
- **In an emergency,** tap **CANCEL**, then **CONFIRM**. The machine stops and the door unlocks (on the dryers after a short cool-down).

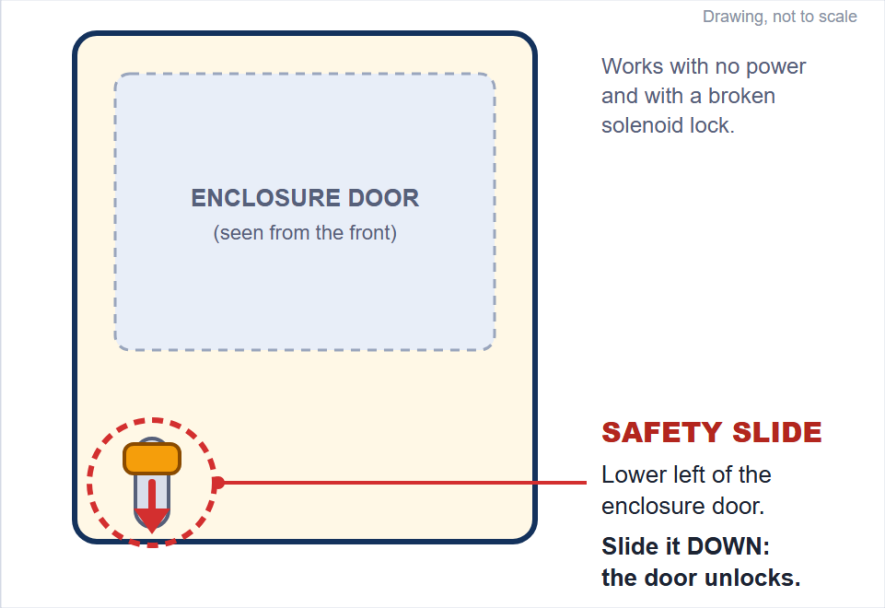
If your helmet is stuck after a power cut or a fault, do not pull on the door. Slide the **safety lock down** (lower left of the enclosure door): the door unlocks at once. Then tell the staff.

If the door will not open

The door is locked on purpose while a machine runs. If the screen does not answer, the power is off, or the lock is broken, you can always open the door yourself with the **safety slide**.

1. Look at the **lower left corner of the enclosure door**. The safety slide is there.

2. Slide it **down**. The door unlocks at once, with or without power.
3. Open the door and take your helmet out. Tell the staff afterwards.



The safety slide: lower left of the enclosure door. Slide it down to unlock (drawing)

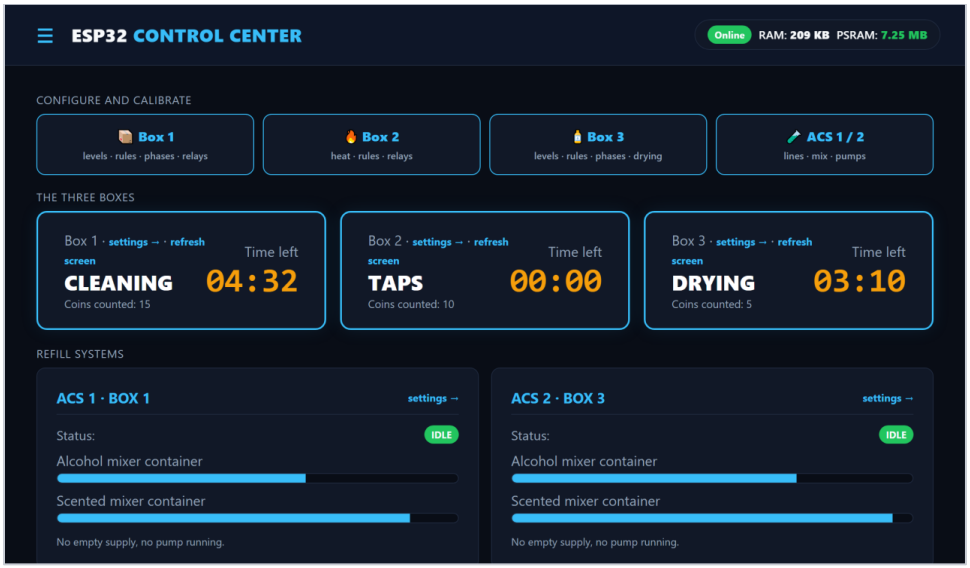
If the machine was running, opening the door stops it (SAFETY PAUSE) and switches the UV light and the heat off. The UV light is strong: do not look into the chamber until it is off. The helmet may be warm.

Questions

Question	Answer
The screen went white.	Wait a moment, it restarts itself. If it stays white, tell the staff.
I inserted a coin but nothing changed.	Insert the coin again slowly. Tell the staff if the count does not move.
Can I get a refund?	Coins are not refunded after the run has started or been cancelled.
The time is too long / too short.	On the number pad, type the time you want before tapping OK.
Why was I offered a free retry?	Because the helmet may need one more round. It is offered once per visit.
Which station should I use?	Smelly: Box 1 or Box 3 CLEANING. Wet: Box 2 or Box 3 DRYING.
The door will not open.	Wait until the run ends, or tap CANCEL. If the screen does not answer or the power is off, slide the safety lock down (lower left of the enclosure door).

Admin guide (Control Center)

The Control Center: settings, calibration and maintenance



WEB PANEL FOR ADMINS AND TECHNICIANS

Connecting

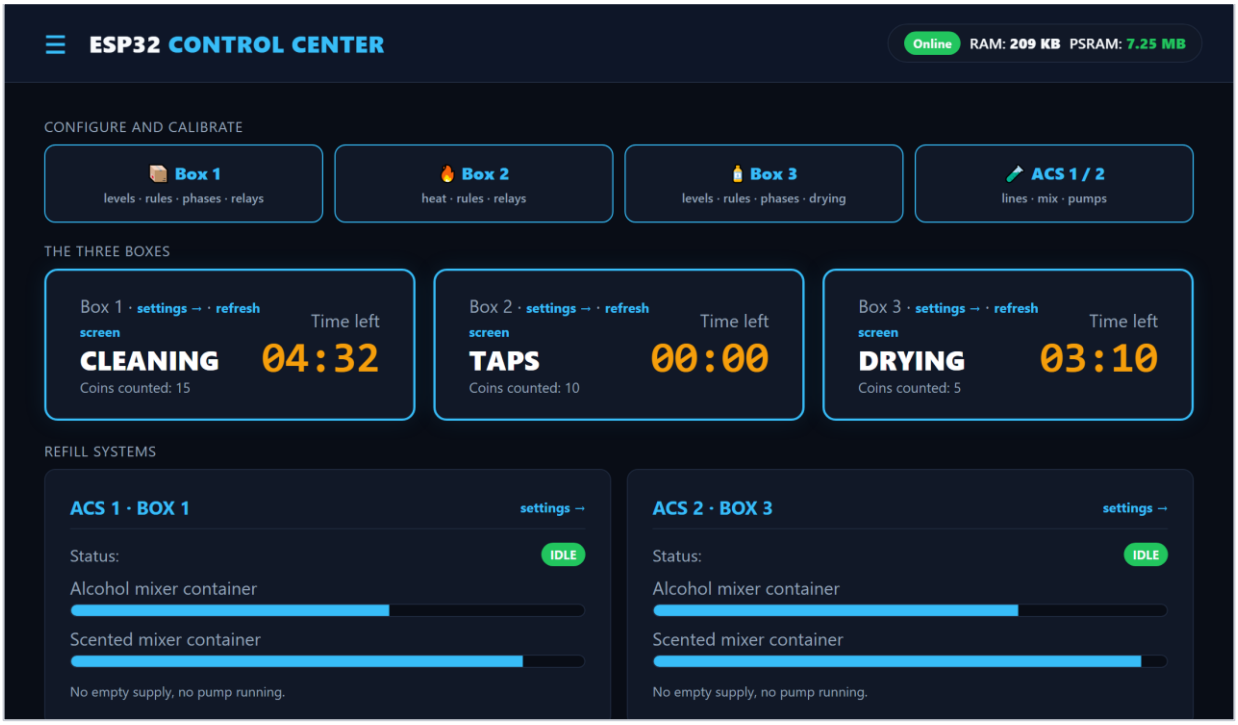
The Control Center is a small web server (the **LocalServer**) inside the machine. You connect to it with a phone or laptop. It does not need internet.

1. Switch on the machine and wait about 30 seconds.
2. On your phone, join the Wi-Fi `ESP32_LOCAL`, password `12345678`.
3. Open a browser and go to `http://192.168.4.1` (the page often opens by itself).
4. The Home page appears. Use the **menu button** (three lines, top left) to open the settings pages.

> The machines keep running and keep every setting even when the LocalServer is off or being replaced. It only shows, and sends your changes.

> The Wi-Fi password is printed here for admins. Change it in the LocalServer program if this manual is shared outside the team.

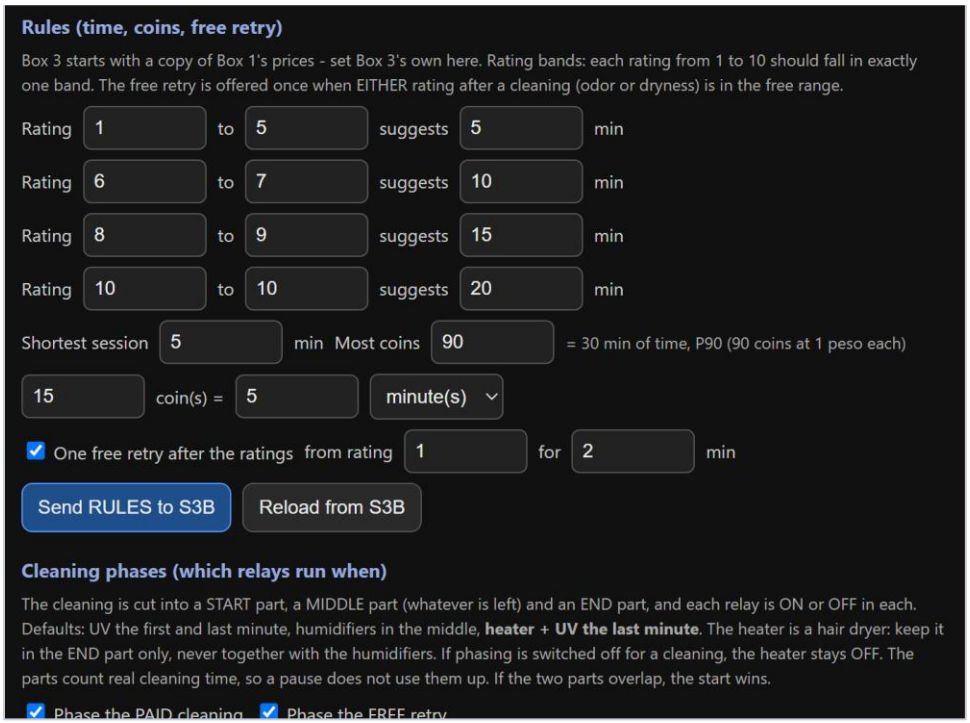
Home page



- **Alert banner** (red): empty supplies, low mixers, a station out of order.
- **Configure and calibrate:** big buttons to Box 1, Box 2, Box 3 and ACS.
- **The three boxes:** current step, coins counted, time left, plus **settings** and **refresh screen** links.
- **Refill systems:** ACS status, mixer levels, empty or running pumps.
- **Statistics:** users served, revenue, average time, coins, last 10 cycles.
- **Menu list:** online/offline badge for every board.

Box settings (rules, levels, phases)

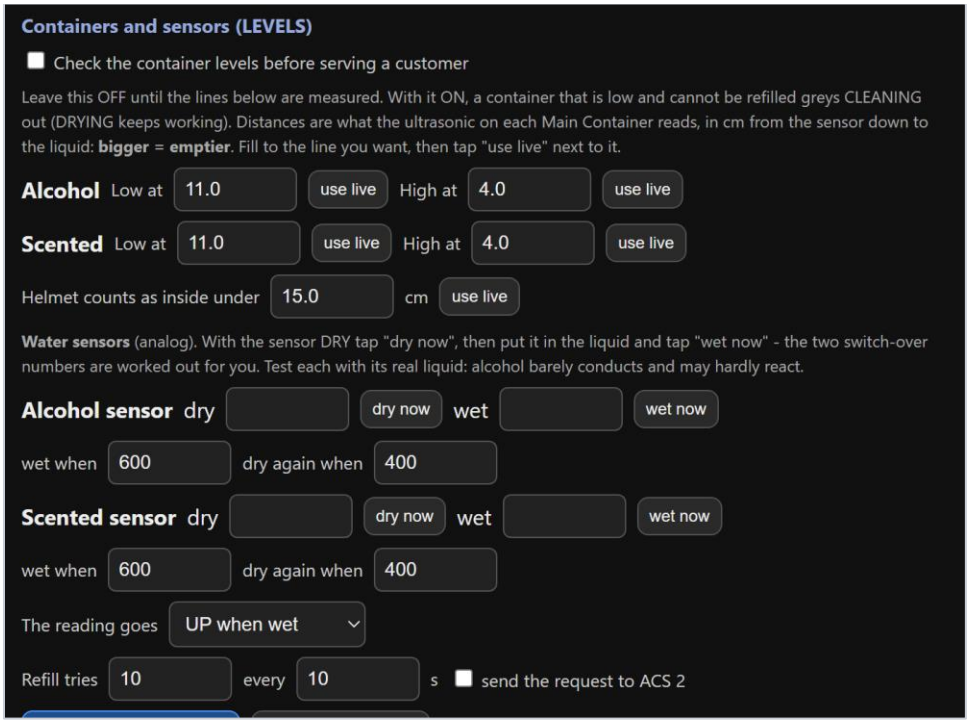
Rules and prices



How to use it (Box 1, Box 2 and Box 3 have the same section):

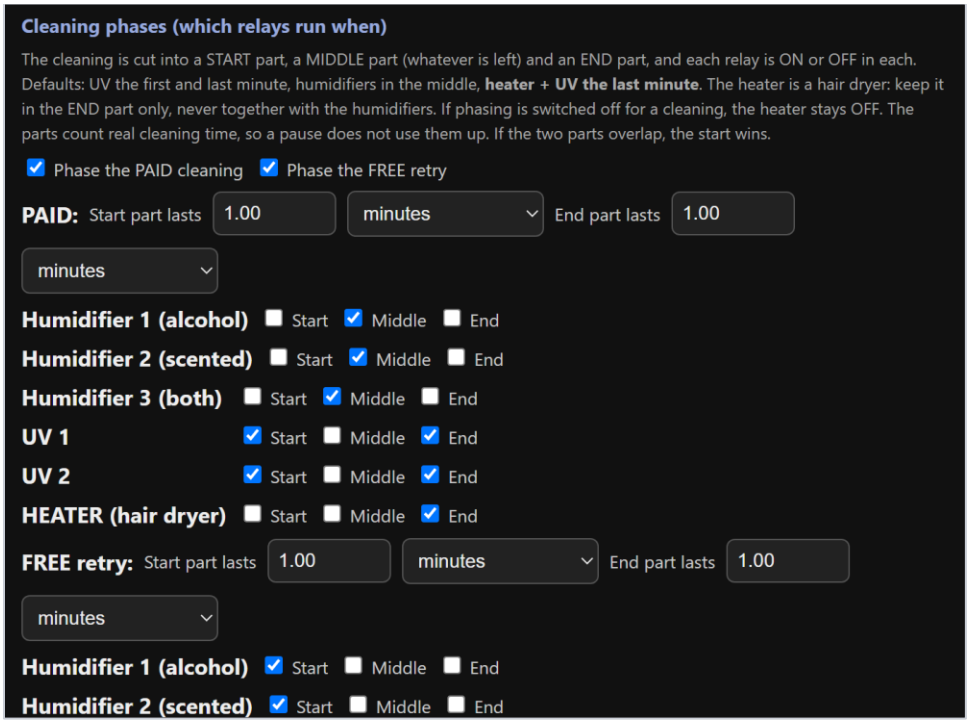
1. Open the box page, scroll to **Rules**.
2. Set the **rating bands** (which ratings suggest how many minutes), the **shortest session**, the **most coins** and the **coin rate** (N coins = X minutes).
3. Set the **free retry** (from which rating, for how many minutes).
4. Tap **Send RULES**. A green tick means the box has saved them.

Containers and sensors



- **Check the container levels:** keep OFF until the numbers are measured.
- **Low / High at:** fill to the line, tap **use live**. Bigger number = emptier.
- **Helmet inside under:** put a helmet in, tap **use live**, raise it a little.
- **Water sensors: dry now**, then **wet now** in the real liquid (alcohol conducts poorly).
- **Refill tries:** tick **send the request to the ACS** only when the ACS works. Then tap **Send LEVELS**.

Cleaning phases



A cleaning is cut into a **START** part, a **MIDDLE** part and an **END** part. Tick when each relay is on.

- Example: UV on in Start and End, mist in the Middle.
- **Always on** = tick Start, Middle and End.
- Box 3: keep the **HEATER** in the End part only, never with a humidifier. The page warns you if they overlap.
- Paid cleaning and the free retry have their own tables.

Tap **Send PHASES**.

Heating and drying

- **Box 2 - Heating:** set the **helmet distance** (put a helmet in, **use live**) and the **cool-down seconds** (default 60, 5 to 900). Tap **Send HEAT**.
- **Box 3 - Drying:** set the **cool-down seconds**, then **Send DRYING**. The drying time itself comes from the customer's choice and the Rules.

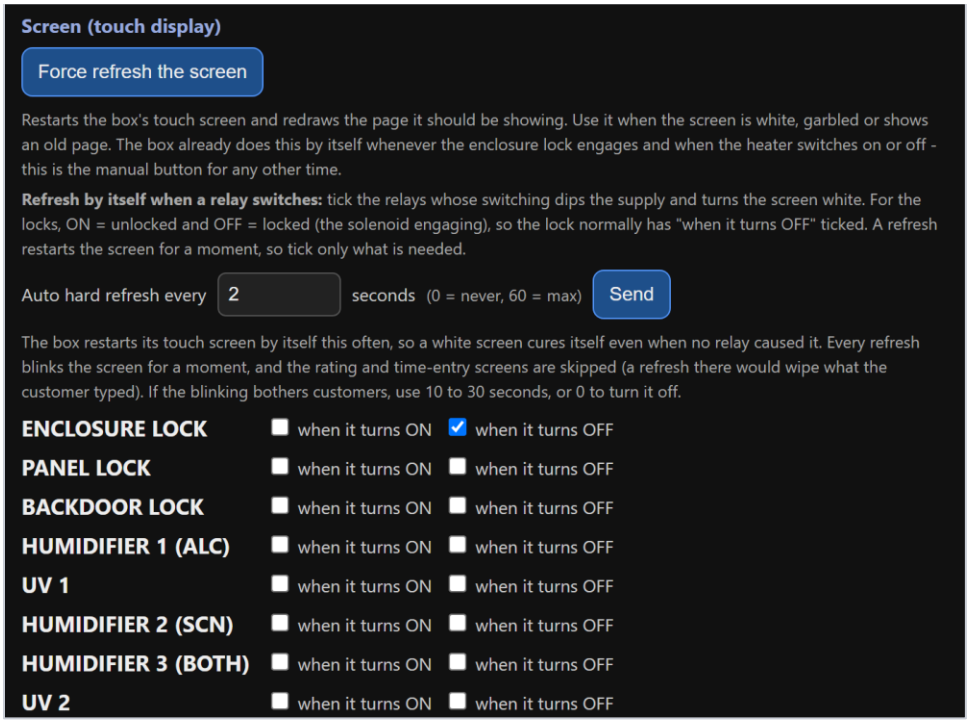
Good to know

- Every setting is saved **inside the box's own board**, so it survives a power cut or a LocalServer replacement.
- If a number is outside its allowed range, the box refuses it. The green tick will not appear.
- Use **Reload** beside a Send button to bring back what the box really has.

> Change one thing at a time and test it with a short session before leaving it.

Screen refresh and maintenance

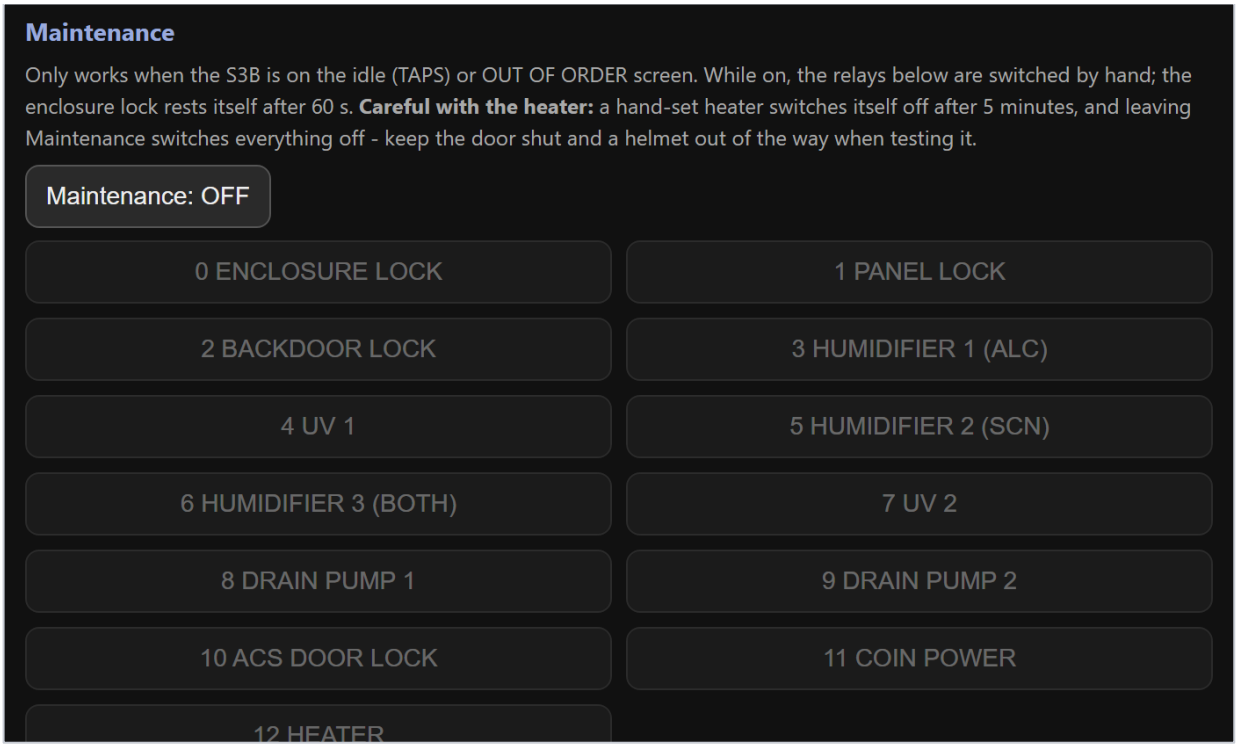
Screen (touch display)



A heavy load switching on can make the touch screen go **white**. Three tools cure it:

- **Force refresh the screen:** restarts it now.
- **Auto hard refresh every N seconds:** restarts it by itself (0 = never, 60 = max). Rating and number-pad screens are skipped.
- **Refresh when a relay switches:** tick the relays (ON and/or OFF) that should trigger it. The lock is normally ticked OFF.

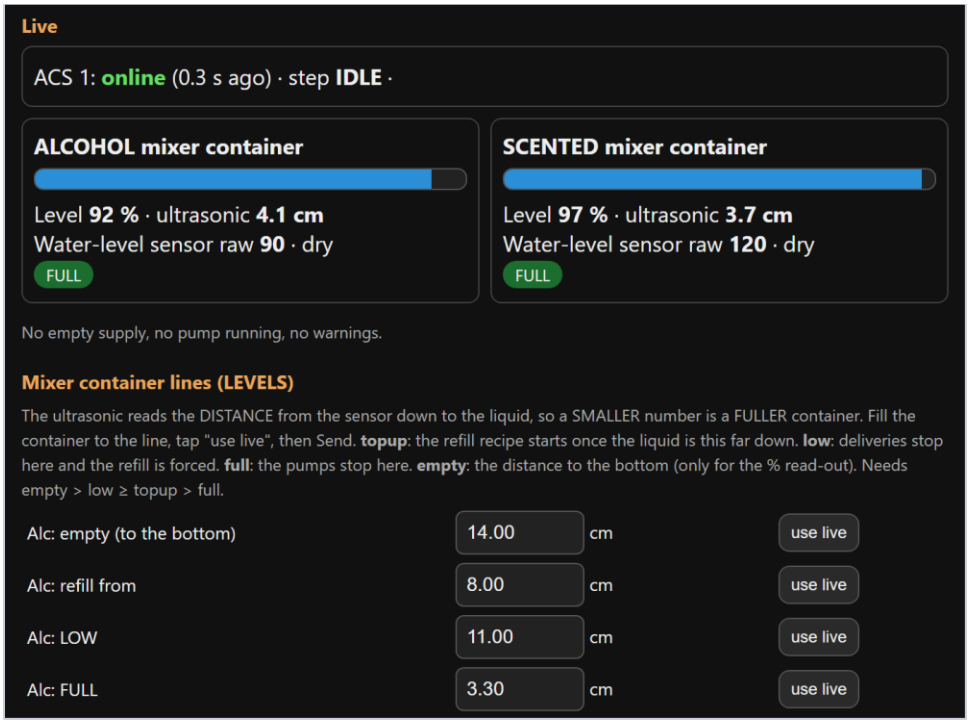
Maintenance mode



1. Make sure the station is on its start screen (TAPS) or OUT OF ORDER.
2. Tap **Maintenance**. The machine shows MAINTENANCE.
3. Tap a relay button to switch it on or off by hand. Green = on.
4. Tap **Maintenance** again. Everything switches off.

> The hand-set heater turns itself off after 5 minutes and the lock after 60 seconds. Keep the door shut and clear when testing the heater.

Refill station (ACS)



ACS 1 refills Box 1, **ACS 2** refills Box 3. Each has three supply tanks (water, alcohol, scented), two mixer containers, and six pumps. Choose the ACS at the top of the page.

- When a mixer drops to its **refill from** line, the ACS adds water first, then the alcohol (or scented liquid) up to FULL.
- When a box asks, the matching **mixer pump** sends one short shot to the box.

Setting up the ACS

1. **Mixer container lines:** fill each mixer to the line and tap **use live** for *refill from*, *LOW*, *FULL* and *empty*. Smaller number = fuller. Tap **Send LEVELS**.
2. **Mix:** the water share of each top-up (30% water = 70/30 alcohol). Tap **Send MIX**.
3. **Timing / water-level sensors:** use **dry now / wet now** for each sensor and **Work out the thresholds**. Tap **Send TIMING**.
4. **Check the pumps** one at a time in Maintenance before the first automatic run.

> Use plain water in all three supply tanks for the first test.

Reading the ACS

You see	It means
IDLE	Nothing to do. Mixers are above the refill line.

You see	It means
ALC WATER IN ALC ALCOHOL IN	Filling the alcohol mixer: water, then alcohol.
SCT WATER IN SCT SCENTED IN	Filling the scented mixer.
DELIVER ...	Sending one shot to the box.
WATER / ALCOHOL / SCENTED EMPTY	That supply tank is empty (or its pump is dead). The pump stopped and it retries in 5 minutes.
Half-made mix	A recipe was interrupted. The mixer is not delivered until it is finished.

Try it and fix it

- **Ask: alcohol / scented** pretends to be the box and makes one delivery shot. **Stop** cancels it.
- **Retry now** forgets an EMPTY verdict after you refilled a tank.
- **Mixers are fine as they are** clears the half-made flag when the mixers really hold good mix.
- **Maintenance:** run each of the six pumps by hand, one at a time. Each pump stops itself after 60 seconds.

Refilling a supply tank

1. The three supply tanks have no sensors: look at them during your daily check.
2. Refill the tank that is low or reported EMPTY.
3. Tap **Retry now**.

> Never run a pump dry. If liquid does not move when a pump runs, check the tube and the relay wiring first.

Troubleshooting and care

Troubleshooting

Problem	What to do
Screen white or garbled	Box page, **Force refresh the screen**. If it returns, tick the relay that causes it or lower the auto-refresh time.
A board shows OFFLINE	Check its power and wiring, then watch its badge on Home.
Station shows OUT OF ORDER	One board is silent (sensor or relay board). See the offline badges.
CLEANING is UNAVAILABLE	Containers are low. Check the ACS; refill, then **Retry now**.
Coins not counted	Use the coin check on the box page: pulses should rise by the coin's value.
Touch is off or upside down	The terminal needs its touch calibration or rotation checked.
Home says last restart BROWNOUT	Power dipped. Use a stronger supply and a short cable.
Door will not open, helmet stuck	Slide the safety lock down (lower left of the enclosure door): it unlocks the door at once, with or without power. Then find the cause: box page, Maintenance, ENCLOSURE LOCK (Fix it: The door will not open).

Care and spare parts

Daily check

- Home page: every badge **ONLINE**, no red banner.
- Look at the three ACS supply tanks (they have no sensors).
- Check the revenue and the number of users.

When a board breaks

1. Keep one spare ESP32 per role, **flashed and labelled** (A1, S3B, ACS 2...).
2. Swap it in and wire it exactly like the old one.
3. Watch its badge turn ONLINE on Home, then re-enter that box's settings if they were lost.
4. After rewiring, test each relay or pump by hand in Maintenance first.

> S3A, S3B, S3C and the LocalServer are ESP32-S3 boards (16 MB, OPI PSRAM). The relay boards A3 and B3 must be plain ESP32 boards.

Online page (Monitor, Alerts, Reports)

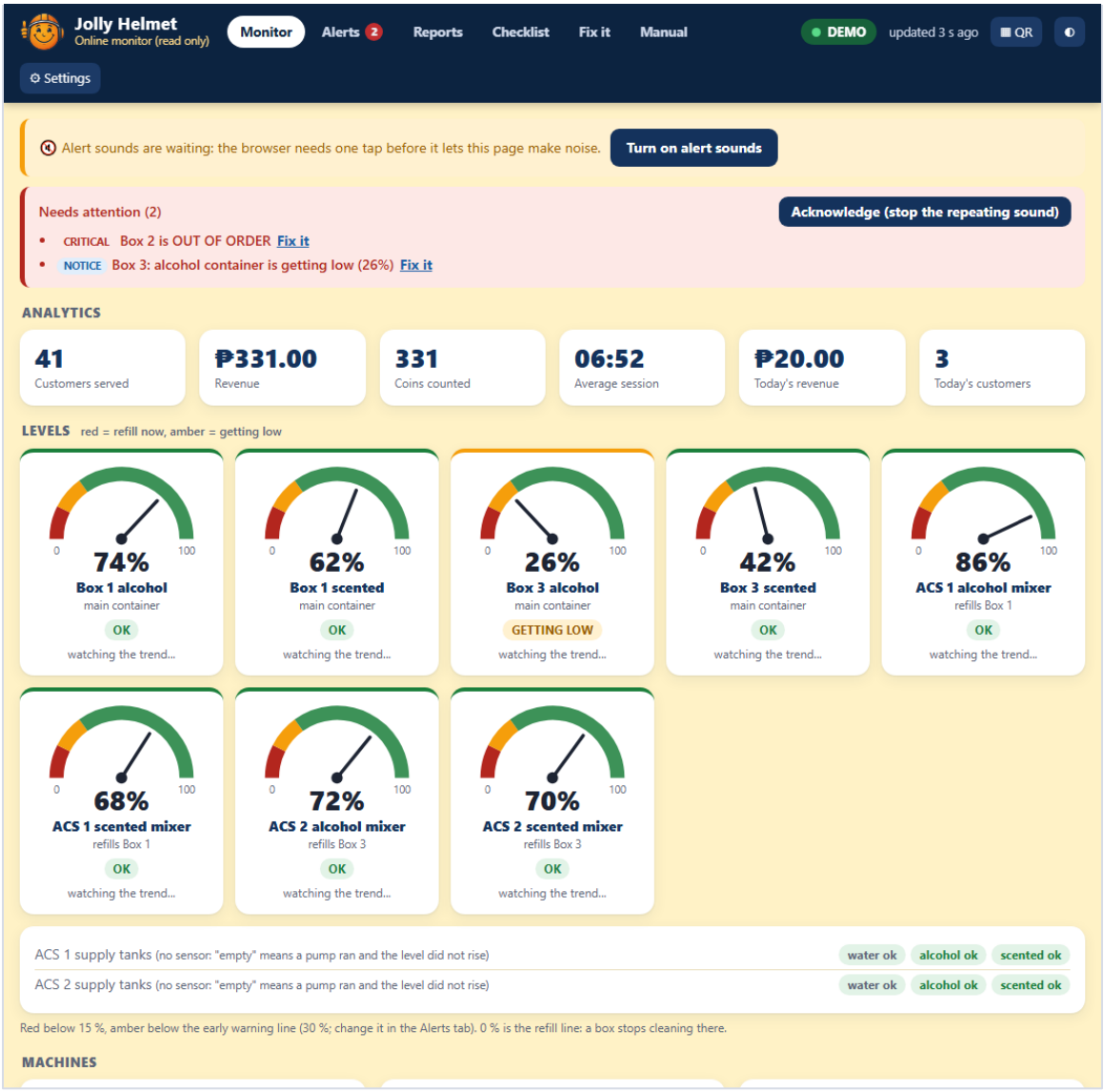
The **online page** shows your machines on a phone or a PC, wherever you are. It only **watches**: nothing on it can start, stop or change a machine. Open it with the address your technician gave you. The first time it asks for the Blynk token: paste it once, it stays in that browser.

The tabs

Tab	What it is for
Monitor	The live state of every box and refill station, the level dials, the revenue and the customers served.
Alerts	What needs attention now, a log of what happened, and the sound of each alert.
Reports	Revenue and customers per day, the busiest hours and days, what each box earns. CSV files and a printable report.
Checklist	The daily, weekly and monthly checks. Tap them as you do them.
Fix it	Yes / no questions that lead you to the fix. The same guides as the decision trees in the Reference chapter.
Manual	This manual. The customer chapters are open to everybody. The admin, technician and reference chapters open with the **team password**.
QR button	The QR code and a printable sign for the customer page. Stick the sign on every machine.

Level dials

Every container has a dial. **0** is the refill line: a box stops cleaning there. The dial is green when the container is fine, **amber** from the early warning line (30 %) and **red** under 15 %. So you can refill **before** a box shows OUT OF ORDER. Under each dial the page says whether the level is falling and about how long it will last.



The Monitor tab: alerts at the top, the level dials, then the machines (demo data)

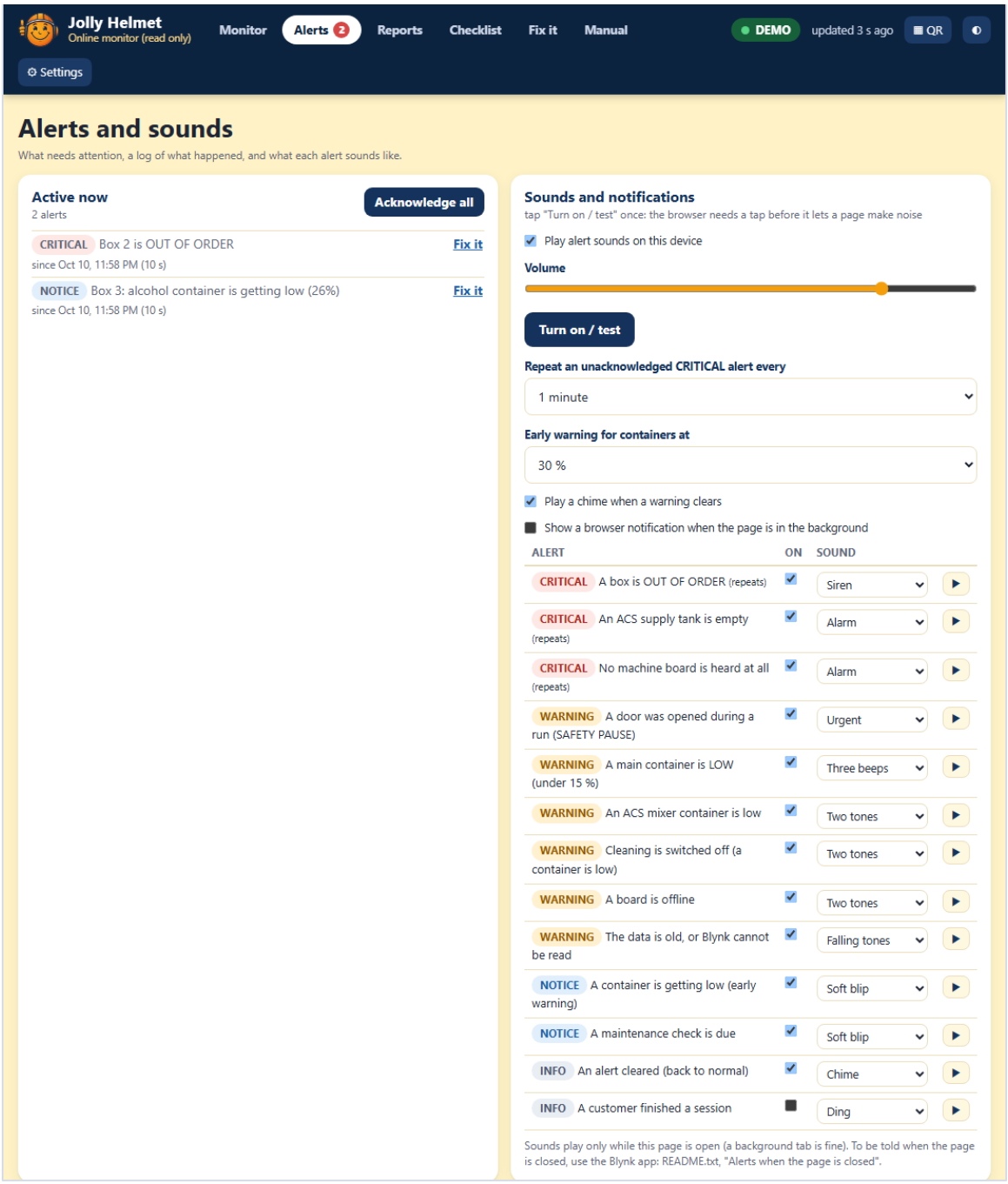
Alerts and sounds

When something needs attention the page shows a banner, puts a number on the tab and the tab icon, and plays a sound. Every kind of alert has its own sound, so you can tell them apart without looking.

Alert	Level	Sound	What to do
A box is OUT OF ORDER	Critical	Siren. Repeats.	Fix it: *A box shows OUT OF ORDER*. Usually a container needs a refill or a board is offline.
An ACS supply tank is empty	Critical	Alarm. Repeats.	Fill that tank, then use Retry now on the ACS page.
No board is heard at all	Critical	Alarm. Repeats.	Check the power and the radio at the site. Fix it: *A board shows OFFLINE*.
A door was opened during a run	Warning	Three rising beeps, twice	The run continues when the door is shut. Find out why it was opened.

Alert	Level	Sound	What to do
A container is low (under 15 %)	Warning	Three beeps	Refill it now.
An ACS mixer is low, cleaning is switched off, a board is offline	Warning	Two tones (ding-dong)	Look at the ACS tanks and the Home page. Use Fix it.
The data is old, or Blynk cannot be read	Warning	Falling tones	Check the internet at the site and the Online Server. Fix it: *The web page says OLD DATA*.
A container is getting low	Notice	Soft blip	Plan a refill.
A maintenance check is overdue	Notice	Soft blip	Open the Checklist tab.
An alert cleared	Info	Chime	Nothing: it is back to normal.

- A browser keeps a page silent until you have tapped it once. Tap **Turn on alert sounds**, then keep the page open (a background tab is fine).
- Press **Acknowledge** to stop a repeating alarm. A new alert still sounds.
- In the Alerts tab you can change the sound of every alert, switch one off, set the volume, change how often a critical alert repeats and move the early warning line.



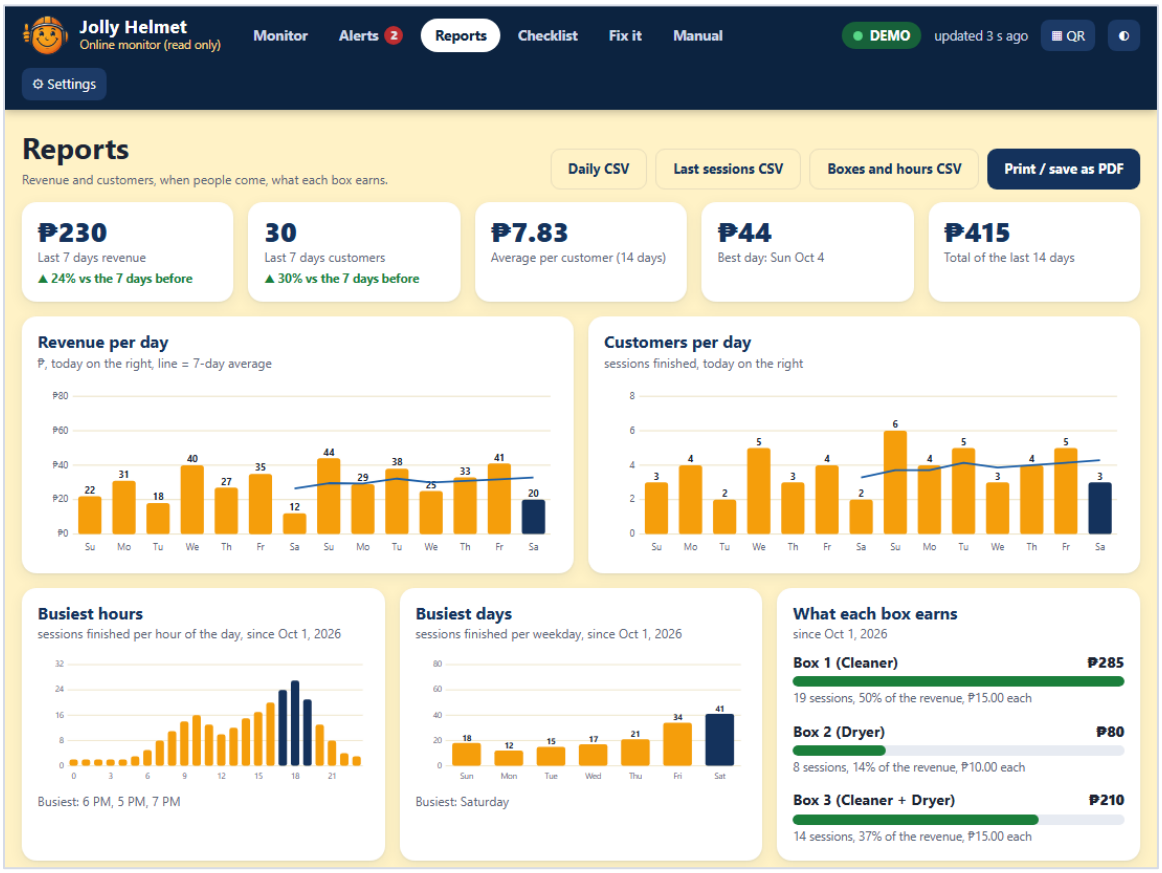
The Alerts tab: active alerts, the sound settings and the log (demo data)

A page that is closed, or a phone that is locked, cannot play a sound. For alerts on a closed page use the Blynk app (Technician guide, Online monitoring, *Alerts when the page is closed*).

Reports

- **Revenue and customers per day** for the last 14 days, with the 7-day average, and this week against the week before.
- **Busiest hours** and **busiest days**: when the customers come.

- **What each box earns**: sessions, revenue and the share of each box.
- **Daily CSV**, **Last sessions CSV** and **Boxes and hours CSV** open in Excel. **Print / save as PDF** makes a clean report page.

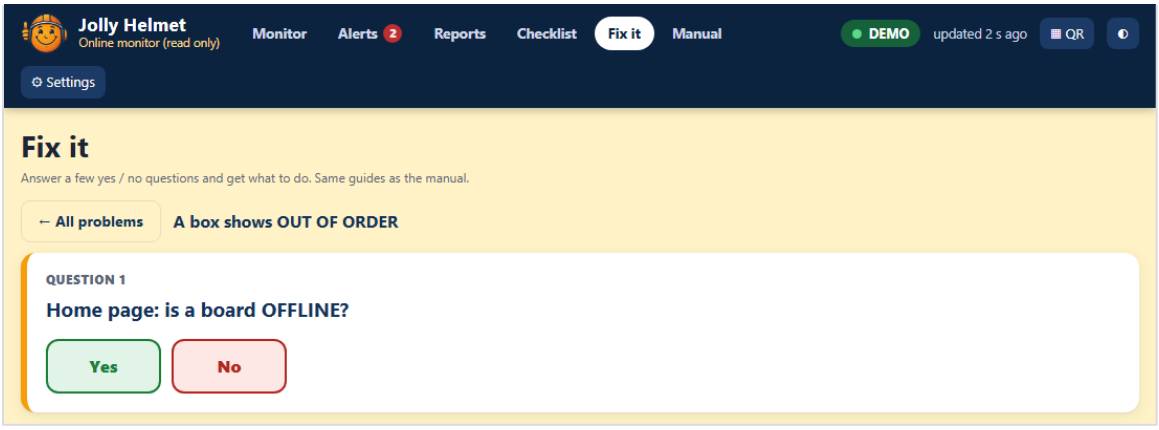


The Reports tab (demo data)

Checklist and Fix it

The **Checklist** follows the maintenance schedule of the Technician guide. Tap a machine under a check when you have done it there. The page shows when each check was last done and what is due. It is saved in the browser you use: use **Save a backup** to move it to another phone.

Fix it asks one yes / no question at a time and ends with what to do. If an alert is on, the matching guide is suggested at the top.



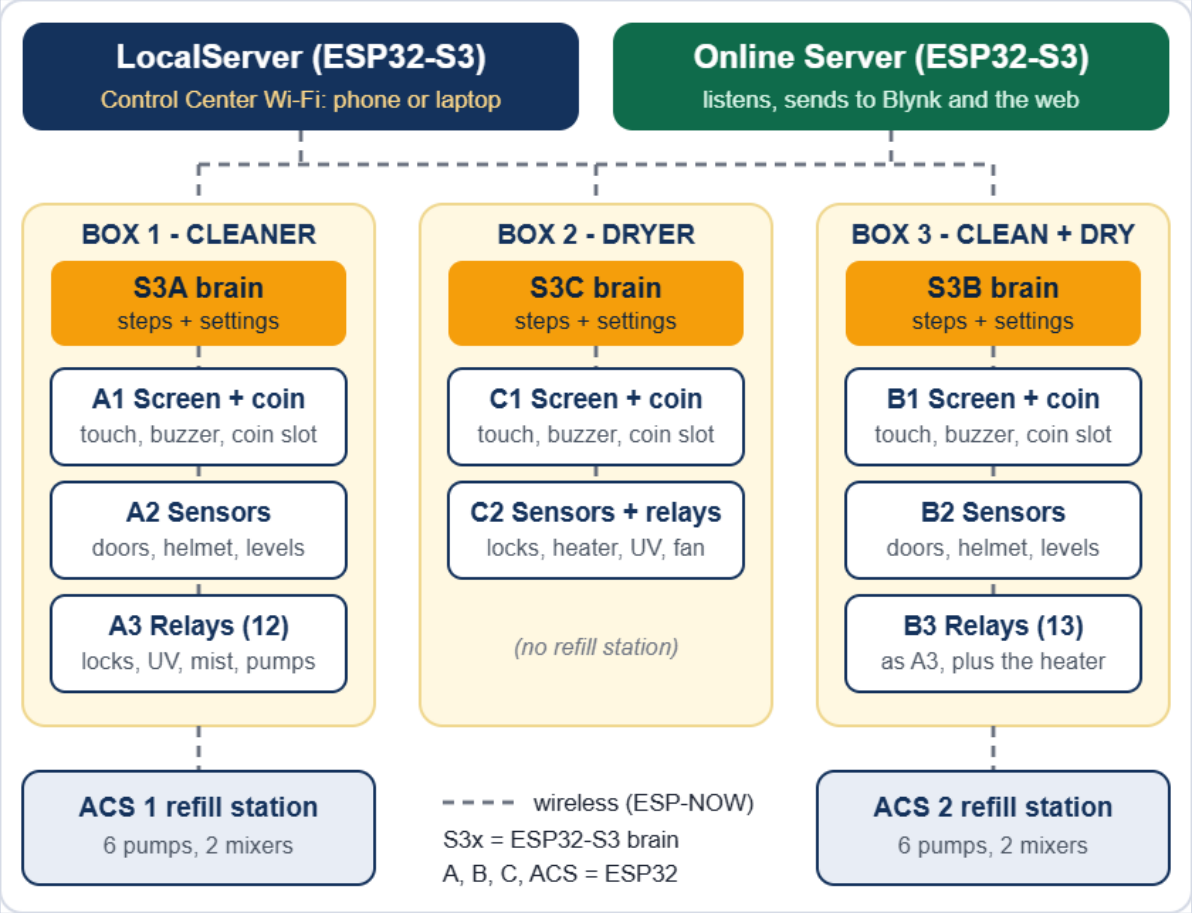
Fix it: one question at a time

Technician guide (Hardware, online monitoring and maintenance)

Components, pinouts, wiring, firmware, online monitoring and maintenance

How it is built

Each machine is a small team of ESP32 boards that talk by radio (ESP-NOW, channel 1). The only wires between the boards are power wires.



The whole installation: three boxes, two refill stations, the LocalServer and the Online Server

The 15 boards

Board	Where	What it does (sketch folder)
S3A	Box 1	Brain. Runs every step and keeps the settings. (ESP32_S3A)
A1	Box 1	Touch screen, coin slot, buzzer. (ESP32_A1)
A2	Box 1	Sensors: 4 doors, helmet, 2 containers, 2 water sensors. (ESP32_A2)
A3	Box 1	12 relays: 4 locks, 3 humidifiers, 2 UV, 2 drain pumps, coin power. (ESP32_A3)
S3C	Box 2	Brain. (ESP32_S3C)
C1	Box 2	Touch screen, coin slot, buzzer. (ESP32_C1)
C2	Box 2	3 door sensors, helmet sensor and 6 relays: 3 locks, heater, UV, fan. (ESP32_C2)
S3B	Box 3	Brain. (ESP32_S3B)

Board	Where	What it does (sketch folder)
B1	Box 3	Touch screen, coin slot, buzzer. (ESP32_B1)
B2	Box 3	Sensors, same as A2. (ESP32_B2)
B3	Box 3	13 relays: same as A3 plus the heater. (ESP32_B3)
ACS 1	Box 1 refill	6 pumps, 2 mixer containers (ultrasonic and water sensor each). (ESP32_ACS1)
ACS 2	Box 3 refill	Same as ACS 1. (ESP32_ACS2)
LocalServer	Control Center	Wi-Fi ESP32_LOCAL, settings pages, revenue. Also broadcasts its revenue totals for the Online Server. (ESP32_LOCALSRV)
Online Server	Online	Listens to the machines and sends a snapshot to Blynk for the web page. Never controls anything. (ESP32_ONLINESRV)

Brains and terminals

- **Brains** (S3A, S3B, S3C, LocalServer, Online Server) are ESP32-S3 boards with 16 MB flash and 8 MB PSRAM. Nothing is wired to them except power.
- **Terminals** (the A, B, C and ACS boards) are plain ESP32 (WROOM) boards on screw-terminal expansion boards. They hold every sensor, relay and display wire. A3 and B3 must not be WROVER (PSRAM) boards, because GPIO 16 and 17 are relays.
- Settings are saved inside the brains (S3A, S3B, S3C) and the ACS boards. The LocalServer only saves the revenue numbers. The Online Server saves its own Wi-Fi, Blynk token and the daily revenue history.
- If a terminal hears nothing from its brain for 15 seconds, it switches its relays off (heater, UV, mist, pumps, coin power). If a helmet is stuck behind a closed door, it also lets the enclosure lock go.
- The enclosure door also has a mechanical safety slide (lower left) that unlocks it by hand, with no power and no firmware.

Component list

Quantities are for the whole installation: Box 1, Box 2, Box 3, ACS 1 and ACS 2.

Boards and display

Item	Qty	Used in
ESP32 DevKit (ESP32-WROOM-32) with screw-terminal expansion board	10	A1 A2 A3 B1 B2 B3 C1 C2 ACS 1 ACS 2
ESP32-S3 DevKit N16R8 (16 MB flash, 8 MB OPI PSRAM)	5	S3A S3B S3C LocalServer Online Server
2.8 in SPI TFT with resistive touch (ST7789, 240 x 320)	3	A1, B1, C1
Allan multi-coin acceptor, 12 V	3	One per box
Active buzzer	3	A1, B1, C1
Resistor 10 kΩ	3	Coin signal lines

Sensors

Item	Qty	Used in
MC-38 magnetic door sensor (reed)	11	Box 1: 4, Box 2: 3, Box 3: 4
Ultrasonic distance sensor HC-SR04 (5 V)	11	Helmet: 3 (one per box). Main containers: 4. ACS mixers: 4.
Analog water-level sensor	8	Box 1: 2, Box 3: 2, ACS 1: 2, ACS 2: 2

Relay boards

Item	Qty	Used in
16-channel relay board, active-LOW	2	A3 (12 used), B3 (13 used)
6-channel relay module	1	C2 (all 6 used)
8-channel relay module with JD-VCC	2	ACS 1, ACS 2 (6 used each)

Loads

Item	Qty	Used in
12 V solenoid door lock	11	Box 1: 4 (enclosure, panel, back, ACS side). Box 2: 3. Box 3: 4.
Humidifier (mist maker), 5 V	6	Box 1: 3, Box 3: 3 (alcohol, scented, both)
Drain pump, 5 V	4	Box 1: 2, Box 3: 2
Refill pump, 5 V	12	ACS 1: 6, ACS 2: 6
UV-C germicidal disinfection lamp, 220 V	5	Box 1: 2, Box 2: 1, Box 3: 2
Heater (hair dryer), 220 V	2	Box 2, Box 3

Item	Qty	Used in
Fan, 5 V	1	Box 2 cool-down

Power and protection

Item	Qty	Used in
12 V power supply	5	Locks, coin slots and the DC-DC modules
DC-DC power module, 12 V in, 3.3 V and 5 V out (multi output)	5	All ESP32 boards, screens, relay boards, sensors, mist makers, pumps
Diode 1N4007	11 or more	One across every lock solenoid
Fuses, 220 V inlet or outlets, earth	-	Mains side: UV-C lamps and heaters

Not electrical: 2 main containers (alcohol, scented) in Box 1 and in Box 3. In each ACS: 3 supply tanks (water, alcohol, scented) and 2 mixer containers. Tubing and fittings.

What is in each unit

Item	Box 1	Box 2	Box 3	ACS 1	ACS 2
ESP32 terminal boards	3	2	3	1	1
ESP32-S3 brain	1	1	1	-	-
Touch screen, coin slot, buzzer	1	1	1	-	-
Door sensors	4	3	4	-	-
Ultrasonic sensors	3	1	3	2	2
Water-level sensors	2	-	2	2	2
Door locks	4	3	4	-	-
Humidifiers	3	-	3	-	-
Drain pumps	2	-	2	-	-
Refill pumps	-	-	-	6	6
UV-C lamps (220 V)	2	1	2	-	-
Heater	-	1	1	-	-
Fan	-	1	-	-	-

Pinouts

GPIO numbers are the ESP32 pin names (D4 = GPIO4). Every relay input is active-LOW: the pin goes LOW to switch the relay ON. These tables match the flashed firmware and the pin sheet *RevisedJollyHelmet_Checked.xlsx*.

Relay input numbers (IN1, IN2...) come from the Cirkit drawing. The GPIO is what the firmware drives. If a relay sits on a different input, trust the GPIO and fix the label.

S3A, S3B, S3C and LocalServer - nothing to wire

These four boards have no external wiring: power (5 V) and the USB cable for flashing only. GPIO 35, 36 and 37 are used inside the module by the OPI PSRAM: never connect anything to them.

A1, B1, C1 - screen, coin slot, buzzer

Identical wiring on all three boards:

GPIO	Signal	Wired to
15	TFT CS	Display CS
4	TFT RESET	Display RST
2	TFT DC	Display DC (RS)
13	MOSI (shared)	Display SDI (SDA) and touch T_DIN
14	SCK (shared)	Display SCK (SCL) and touch T_CLK
27	Touch CS	Touch T_CS
12	Touch MISO	Touch T_DO
23	Coin signal	Allan COIN, through a 10 kΩ resistor
21	Buzzer	Active buzzer +
5V	Display power	Display VCC and LED (BLK)
GND	Ground	Display, coin acceptor, buzzer -

- Not connected: display SDO (MISO), touch T_IRQ, coin COUNTER.
- Coin acceptor 12 V: from the COIN POWER relay on A3 and B3 (Box 1, Box 3). Straight from the 12 V supply in Box 2.
- Free pins: 5, 16, 17, 18, 19, 22, 25, 26, 32, 33.

A2, B2 - sensors (Box 1, Box 3)

GPIO	Signal	Wired to
26	Helmet ultrasonic TRIG	Sensor TRIG
25	Helmet ultrasonic ECHO	Sensor ECHO
32	Alcohol container TRIG	Sensor TRIG

GPIO	Signal	Wired to
33	Alcohol container ECHO	Sensor ECHO
23	Scented container TRIG	Sensor TRIG
22	Scented container ECHO	Sensor ECHO
27	Enclosure door	MC-38 sensor S
14	Panel door	MC-38 sensor S
19	Back door	MC-38 sensor S
18	ACS side door	MC-38 sensor S
34	Alcohol water sensor	SIG (analog, input only)
35	Scented water sensor	SIG (analog, input only)

- Ultrasonic VCC 5 V. Water sensors 3.3 V. Door sensors: the other wire goes to GND, the ESP32 supplies the pull-up (closed = LOW).
- Free pins: 2, 4, 12, 13, 15, 16, 21.

A3 - Box 1 relays (16-channel board)

GPIO	Input	Load	Supply
4	IN1	Enclosure door lock	12 V
16	IN2	Panel door lock	12 V
17	IN3	Back door lock	12 V
23	IN4	ACS side-door lock	12 V
18	IN5	Humidifier 1 (alcohol)	5 V
19	IN6	Humidifier 2 (scented)	5 V
25	IN7	Humidifier 3 (both)	5 V
22	IN9	Coin power (the coin slot's 12 V)	12 V
27	IN10	UV-C lamp 1	220 V
21	IN11	UV-C lamp 2	220 V
33	IN12	Drain pump 1 (alcohol container)	5 V
32	IN13	Drain pump 2 (scented container)	5 V

- Supply column: what each load runs on. The UV-C lamps are 220 V: the relay contact switches the live wire (see **Switching 220 V loads**). Locks and the coin slot are 12 V, humidifiers and drain pumps 5 V.
- IN8 and IN14 to IN16 are not used. GPIO 2, 12 and 15 are left idle (boot pins).

B3 - Box 3 relays (16-channel board)

Pin for pin the same as A3, plus one more relay:

GPIO	Input	Load	Supply
26	IN14	Heater (hair dryer)	220 V via the relay contacts

C2 - Box 2 sensors and relays

GPIO	Signal	Wired to
26	Helmet ultrasonic TRIG	Sensor TRIG
25	Helmet ultrasonic ECHO	Sensor ECHO
27	Enclosure door	MC-38 sensor S
14	Panel door	MC-38 sensor S
19	Back door	MC-38 sensor S

Relays (6-channel module):

GPIO	Input	Load	Supply
4	IN1	Enclosure door lock	12 V
16	IN2	Panel door lock	12 V
17	IN3	Back door lock	12 V
18	IN4	Heater (hair dryer)	220 V
23	IN5	Fan	5 V
5	IN6	UV-C lamp	220 V

- GPIO5 gives a short pulse when the board powers up, so the UV relay may click once. Recommended: when you next reflash C2, move the UV relay to a boot-safe pin (13, 21, 22, 32 or 33) so a 220 V UV-C lamp can never flash at power-up.

ACS 1 and ACS 2 - refill stations

Identical wiring on both. Relays (8-channel module with JD-VCC, 6 used):

GPIO	Input	Pump	Supply
18	IN1	Alcohol pump (tank to alcohol mixer)	5 V
19	IN2	Scented pump (tank to scented mixer)	5 V
21	IN3	Water pump for the alcohol mixer	5 V
23	IN4	Water pump for the scented mixer	5 V
22	IN5	Mixer alcohol pump (mixer to the box)	5 V
25	IN6	Mixer scented pump (mixer to the box)	5 V

Sensors:

GPIO	Signal	Wired to
4	Alcohol mixer ultrasonic TRIG	Sensor TRIG

GPIO	Signal	Wired to
14	Alcohol mixer ultrasonic ECHO	Sensor ECHO
26	Scented mixer ultrasonic TRIG	Sensor TRIG
33	Scented mixer ultrasonic ECHO	Sensor ECHO
34	Alcohol mixer water sensor	SIG (analog)
35	Scented mixer water sensor	SIG (analog)

- The three supply tanks (water, alcohol, scented) have no sensors. The ACS detects an empty tank when the mixer level does not rise.
- The pumps are 5 V. Check each pump by hand in Maintenance before the first automatic run (the relay inputs were mixed up once).

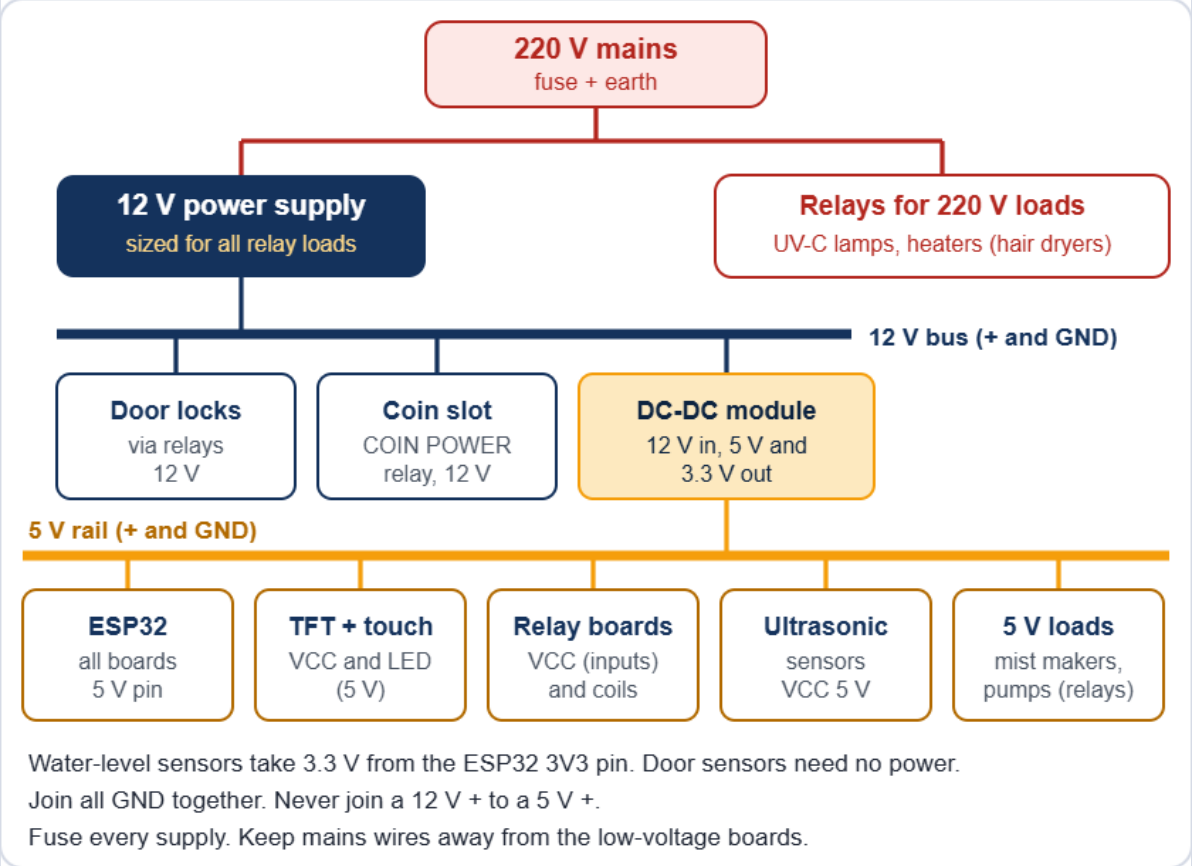
Circuit and wiring

Circuit summary

- Every box and refill station is a group of ESP32 boards. They share only power and the radio channel.
- Every lock, lamp, pump and heater is switched by a relay contact. Every sensor is read directly by a terminal board.
- Voltages: 12 V (door locks, coin slots), 5 V (ESP32 boards, screens, relay boards, HC-SR04 ultrasonic sensors, humidifiers, pumps), 3.3 V (water-level sensors, from the ESP32 3V3 pin or the DC-DC module), 220 V (the UV-C lamps and the heaters).

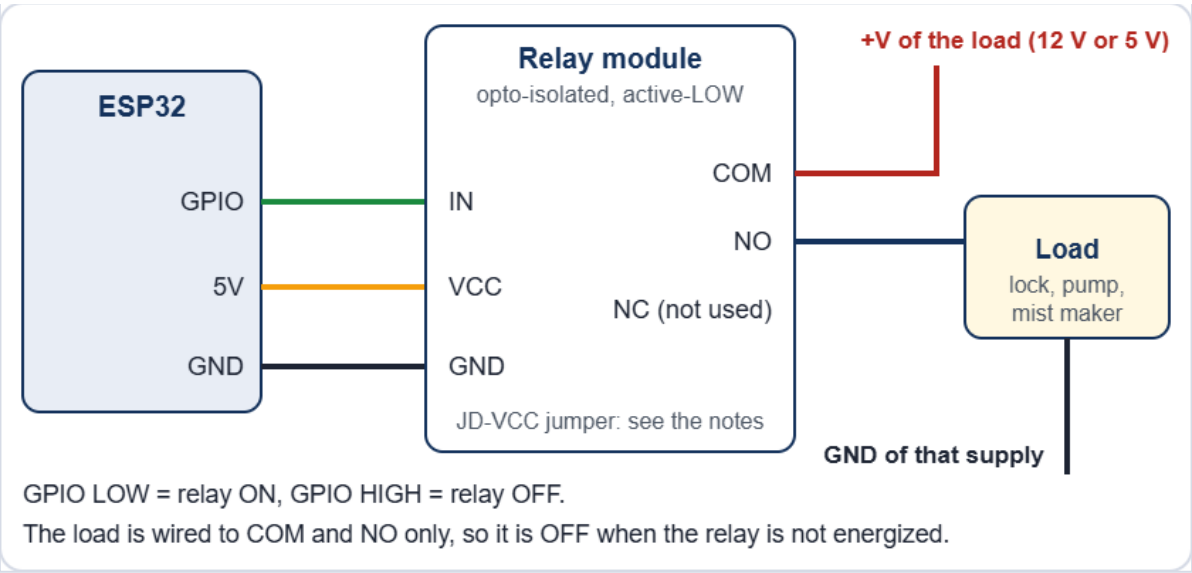
Power and grounding

- Measure every output of the DC-DC module (5 V and 3.3 V) with a meter before you connect boards. Never put 12 V on an ESP32 pin or on the 5 V rail.
- Join all grounds: 12 V -, 5 V - and every ESP32 GND. Never join a 12 V + to a 5 V +.
- A relay module has a JD-VCC jumper. Jumper on: the coils run from the module's VCC (one 5 V supply, shared ground). Jumper off: feed JD-VCC and its GND from a separate 5 V supply.
- Locks, pumps and heaters starting can pull the 5 V down. Symptoms: white screen, LocalServer restarts, odd resets. Cures: short thick 5 V wires, a 470 to 1000 µF capacitor next to the TFT and the ESP32, a separate 5 V for the relay coils. The firmware also switches relays on 25 ms apart.
- Put a fuse on every supply. Keep the 220 V wiring in its own enclosure, away from the boards.



Power for one box (the exact rails are listed in the component tables)

Relay outputs



One relay channel

- Wire the load to COM and NO. Then it stays off when the relay or the board has no power. Leave unused channels unwired.

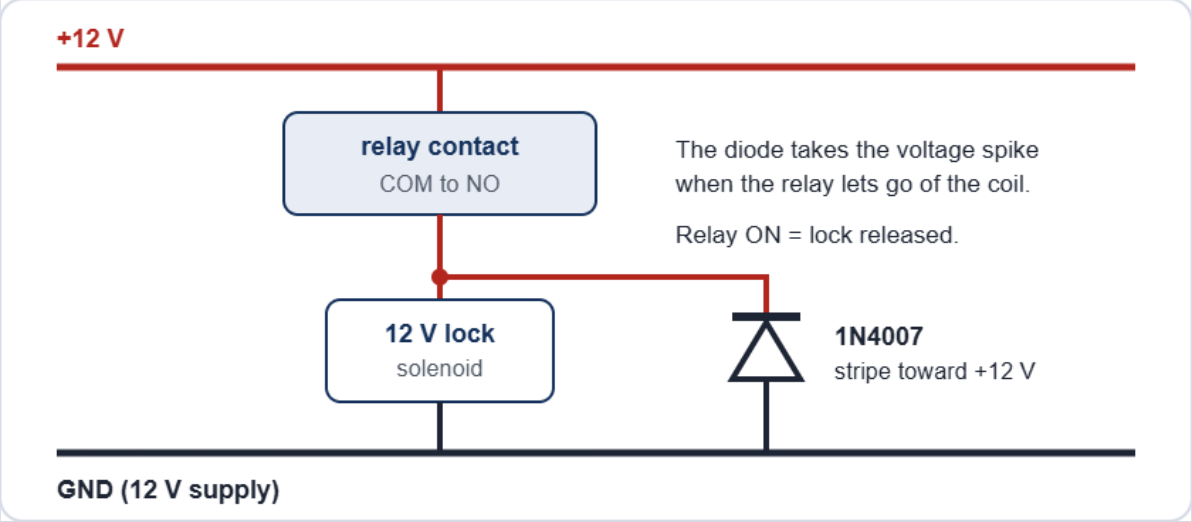
- Relay pins avoid GPIO 0, 1, 2, 3, 5, 12, 14, 15 because they pulse at boot. Safe output pins: 4, 13, 16 to 19, 21 to 23, 25 to 27, 32, 33.
- GPIO 34, 35, 36 and 39 are input only.

Switching 220 V loads (UV-C lamps and heaters)

- The UV-C germicidal lamps and the heaters (hair dryers) run on 220 V. The relay contact switches only the **live** wire. The neutral goes straight from the supply to the load. The relay board's low-voltage side (IN pins, VCC) stays on the ESP32 side.
- Use relay contacts rated for the load. Most relay boards are 10 A at 250 V AC. A hair dryer can draw 8 A or more (read its label). A UV-C lamp draws much less, but its ballast or driver can pull a high current at switch-on.
- Fuse the 220 V feed of each box. Connect the earth wire to every metal part. Use insulated terminal blocks, strain relief on every cable, and keep 220 V wires away from the low-voltage boards and the screen.
- Recommended: a door switch in series with the live wire of each UV-C lamp, so the lamp cannot be lit with a door open even if the firmware or a relay fails.

Never open a box with the 220 V connected. Unplug it first.

Door locks



A door lock with its flyback diode

The locks are **power-to-unlock**: the relay is ON only while the door may open (the OPEN steps, Maintenance, or the trapped-helmet release). With no power a solenoid keeps its door locked, but the enclosure door also has a mechanical **safety slide** that opens it by hand (next block). The 1N4007 takes the voltage spike when the relay lets go of the coil. Without it the spike can reset the board and burn the relay contacts.

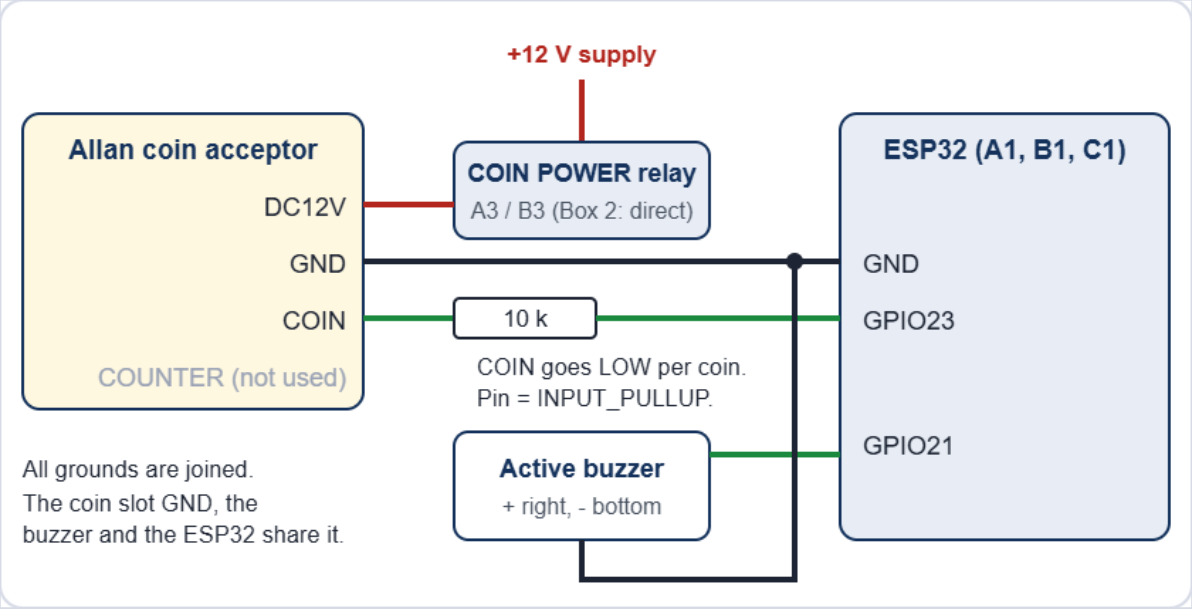
After a power cut the solenoid locks stay shut, but the enclosure door can be opened at once with the **safety slide** (lower left of the door, slide it down). When power is back, release a lock from the Control Center: Box page, Maintenance, ENCLOSURE LOCK. It switches itself off after 60 seconds.

Safety slide on the enclosure door

The enclosure door has a mechanical **safety slide** at its lower left. Sliding it **down** unlocks the door by hand. It works with no power, with a broken solenoid and without any help from the boards. It is the way out for a trapped helmet.

- The slide is part of the door. It is not wired and the firmware does not read it. When the door opens, the door sensor reports it: during a run the box shows SAFETY PAUSE and the UV-C, the mist and the heater are switched off within about 2 seconds.
- A door switch in series with each UV-C lamp (recommended, see Switching 220 V loads) cuts the lamp at once when the door opens, even before the firmware reacts.
- After the slide has been used, put it back in its normal position, close the door and check that the lock holds it. The Maintenance page of the Control Center switches the lock for a test.
- Test the slide every month (see the schedule in Maintenance): with the power off, sliding it down must unlock the door by hand. Never block, tape or remove the slide.

Coin slot and buzzer



Coin acceptor and buzzer on A1, B1 and C1

- The Allan COIN output pulls to ground once per coin pulse. The pin is INPUT_PULLUP and the firmware counts falling edges, so the pin never sees more than 3.3 V. Check it once with a meter on every new unit.
- FAST, MED, SLOW and SET are switches on the acceptor itself, not wires. COUNTER is not used.
- The buzzer is the active type: GPIO21 HIGH = sound. If it sounds weak, or the board resets when it sounds, drive it through a transistor.

Sensors

Sensor	Wire	Goes to
Ultrasonic HC-SR04	VCC	5 V
	TRIG	GPIO (output)

Sensor	Wire	Goes to
	ECHO	GPIO (input), through a divider (the HC-SR04 gives 5 V)
	GND	GND
Water-level	SIG	GPIO34 or GPIO35 (analog, input only)
	VCC	3.3 V
	GND	GND
MC-38 door	S	GPIO (INPUT_PULLUP). Magnet near = closed = LOW
	GND	GND

The HC-SR04 sends 5 V on ECHO and the ESP32 pins are not 5 V tolerant. Put a divider on every ECHO: ECHO to 1 kΩ to the GPIO, and 2 kΩ from that GPIO to GND. (A 3.3 V module such as the HC-SR04P needs none.) The pin sheet shows no divider, so check that every sensor has one.

Screen and touch

The display and the touch controller share the SPI lines MOSI (13) and SCK (14). Each has its own chip select (15 and 27). TFT_eSPI drives the display as an ST7789, 240 x 320. B1 is mounted upside down, so it runs setRotation(3).

Mains and UV safety

Mains can kill. The UV-C lamps and the heaters are on 220 V. Unplug the machine and wait before you open a box. Keep all 220 V wiring fused, earthed, in a closed enclosure with strain relief, away from the low-voltage boards.

UV-C light injures eyes and skin, even for a moment. Never look at a lit lamp and never test it with the door open. The firmware switches UV and mist off the moment a door opens, but firmware is not a safety device. Recommended: a door switch in series with the lamp's live wire, so the lamp cannot be on with a door open, and a warning label on every box.

Firmware and flashing

Arduino IDE settings

Board type	Settings
Terminals (A, B, C, ACS)	Board: ESP32 Dev Module. esp32 core 3.3.7. Upload speed 921600. Other options default.
Brains (S3A, S3B, S3C), **LocalServer** and **Online Server**	Board: ESP32S3 Dev Module. USB CDC On Boot: Enabled. CPU 240 MHz. Flash mode QIO 80 MHz. Flash size 16 MB. Partition: 16M Flash (3MB APP/9.9MB FATFS). PSRAM: OPI PSRAM. Upload speed 921600.

Libraries

- TFT_eSPI 2.5.43: only A1, B1, C1 (edit its User_Setup.h, see below).
- ArduinoJson 7.4.3, Async TCP 3.5.0, ESP Async WebServer 3.12.0: only the LocalServer and the Online Server.
- Everything else (WiFi, ESP-NOW, Preferences) comes with the ESP32 core.

TFT_eSPI User_Setup.h

Setting	Value
Driver	ST7789_DRIVER (the only one not commented out)
TFT_WIDTH, TFT_HEIGHT	240, 320
TFT_MISO, TFT_MOSI, TFT_SCLK	12, 13, 14
TFT_CS, TFT_DC, TFT_RST	15, 2, 4
TOUCH_CS	27
SPI_FREQUENCY, SPI_TOUCH_FREQUENCY	20000000, 1000000

Flash a new or replacement board

1. Find the board in the table **The 15 boards** and open its sketch folder in the Arduino IDE.
2. The folder must hold the same **Shared_Common.h** as ESP32_S3A (the master copy). Different copies make boards ignore each other's messages.
3. Choose the settings above, plug in the USB cable and upload.
4. Open the Serial Monitor at 115200 baud to read the board's log.
5. Watch the board's badge turn ONLINE on the Control Center Home page.
6. For a brain (S3) or an ACS board, enter its settings again on its Control Center page. A new board starts with the default values.

Never run a bench or settings sketch (ESP32_S3A_CONFIG_BENCH, ESP32_S3C_CONFIG_BENCH, ESP32_ACS_CONFIG_BENCH, ESP32_A1_WIDGET_TEST) at the same time as the LocalServer or the real brain. They send the same messages and fight over them.

Test sketches

Sketch	Use
ESP32_A3_BENCH_TEST	Tests A3 alone: it pretends to be S3A and A2.
ESP32_B3_RELAY_TEST	Wi-Fi page (BOX 3 B3 TEST, password 12345678) to switch each B3 relay by hand.
ESP32_BUZZER_ULTRASOUND_TEST	Wi-Fi page to test a buzzer and read an ultrasonic sensor.
ESP32_C1_CALIBRATE	Touch calibration: touch the 4 targets, copy the printed calData line into the terminal sketch.

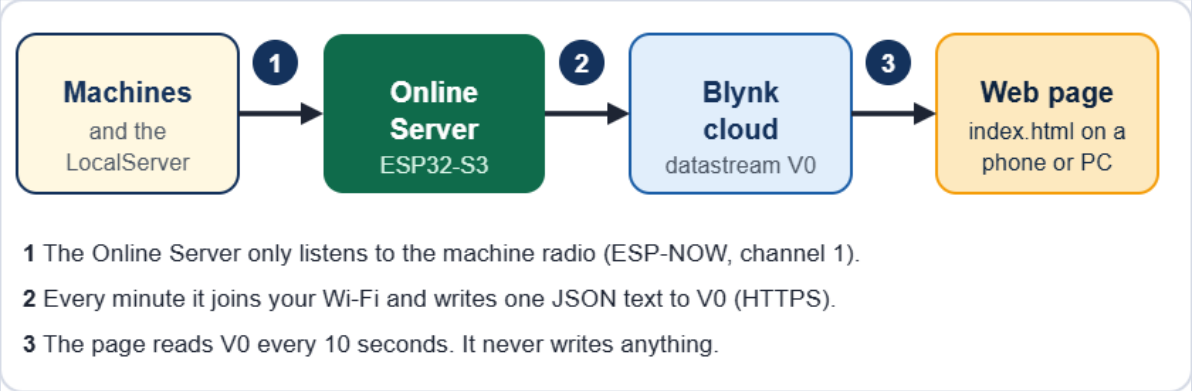
Safety built into the firmware

- Every relay pin is set OFF before it becomes an output, so relays do not pulse at power-up.
- Relays that turn on are staggered 25 ms apart, so loads do not start in the same instant.
- A door that opens during a run switches UV and mist off at once (SAFETY PAUSE).
- A terminal that loses its brain for 15 seconds switches its relays off.
- Hand-set relays in Maintenance switch themselves off: heater after 5 minutes, locks after 60 seconds, ACS pumps after 60 seconds.
- The coin slot only has power while a payment is possible (Box 1 and Box 3).

> The LocalServer must be flashed with the version that broadcasts its revenue totals every 10 seconds, or the Online Server shows no analytics. No other board needs a reflash for the Online Server.

Online monitoring

The **Online Server** is one more ESP32-S3 board. It listens to the machines (it never controls them), sends a snapshot to **Blynk** every minute, and a **web page** shows that snapshot from any phone or PC. Nothing on the web page can start, stop or change a machine.



From the machines to the web page

What you need

- The Online Server (ESP32_ONLINESRV), powered with 5 V, near the machines so it hears every board by radio.
- The LocalServer switched on and flashed with the new version. It sends the revenue totals. Without it the analytics stay empty (the live data still works).
- A 2.4 GHz Wi-Fi network with internet: the school Wi-Fi or a phone hotspot, on any channel.
- A Blynk account and the Blynk Console (blynk.cloud).
- The folder **ONLINE_WEB** (the web page): put it on any web host, or just open index.html from a PC.

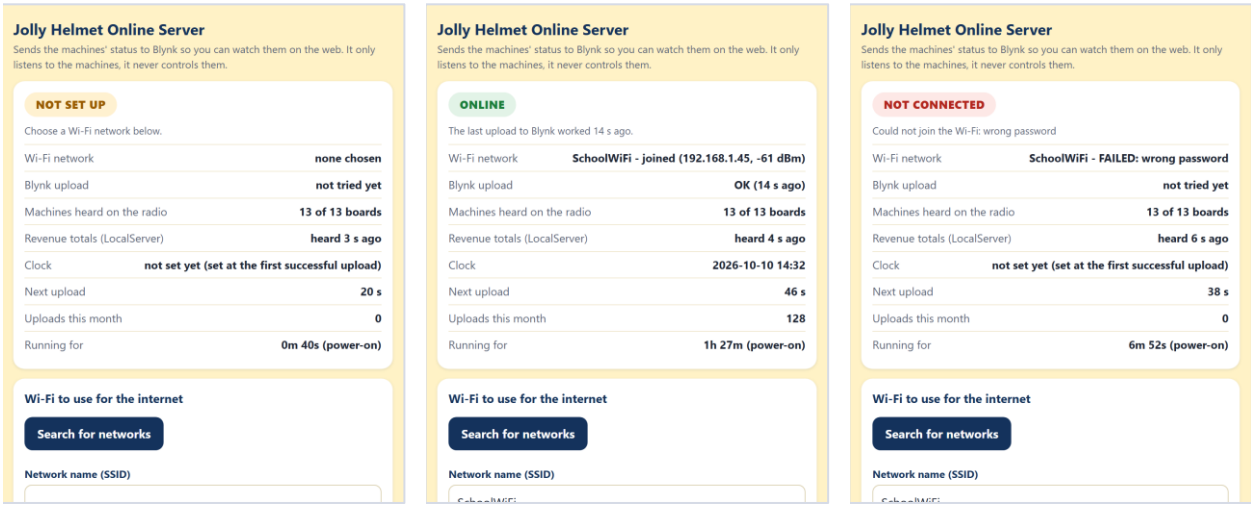
Step 1 - Set up Blynk (once)

1. Sign in to the Blynk Console. Open **Developer Zone**, then **My Templates**, then **New Template**.
2. Name: Jolly Helmet Online Server. Hardware: ESP32. Connection type: Wi-Fi. Click **Done**.
3. Open the **Datastreams** tab. Click **New Datastream**, then **Virtual Pin**. Fill in V0 as in the table below and click **Create**.
4. Optional: make V1 to V4 the same way (single numbers for Blynk's own dashboard and phone app). Default value 0.
5. Optional: open the **Events** tab, click **New Event**, code **alert** (exactly), type Warning or Critical, and switch on a notification (push or e-mail).
6. Click **Save** at the top right of the template. Without Save the device does not see the new datastreams.
7. Open **Devices**, click **New Device**, choose **From template**, pick your template, name it Online Server and click **Create**.
8. Open the device, tab **Device info**, and copy the **Auth Token**. Keep it private.

Name	Pin	Type	Min	Max	Needed?
live	**V0**	String	-	-	**Yes.** The web page reads only this one.
revenue	V1	Double	0	100000000	Optional (2 decimals)
customers	V2	Integer	0	10000000	Optional
alerts	V3	Integer	0	30	Optional
running	V4	Integer	0	3	Optional

Step 2 - Set up the Online Server

1. Flash **ESP32_ONLINESRV** (same settings as the LocalServer, see Firmware and flashing) and power it.
2. On a phone, join the Wi-Fi **JollyHelmet-Online** (password 12345678). The setup page opens by itself. If it does not, open http://192.168.4.1.
3. Tap **Search for networks** and tap your Wi-Fi, or type its name. Type its password.
4. Paste the Blynk **Auth Token**.
5. Choose how often to send. **1 minute** is a good start. Tick "single numbers V1 to V4" and "Blynk event" only if you made them in Blynk. Under **More settings**: the time zone (8 for the Philippines) and, if you want the quota bar on the web page, the number of Blynk messages your plan gives per month.
6. Tap **Save and connect**. The board tests the Wi-Fi and Blynk at once. The setup page may stop answering for a few seconds: this is normal, the board moves to your router's channel. It reconnects by itself and shows the result.



The setup page: not set up yet, online, not connected (wrong password)

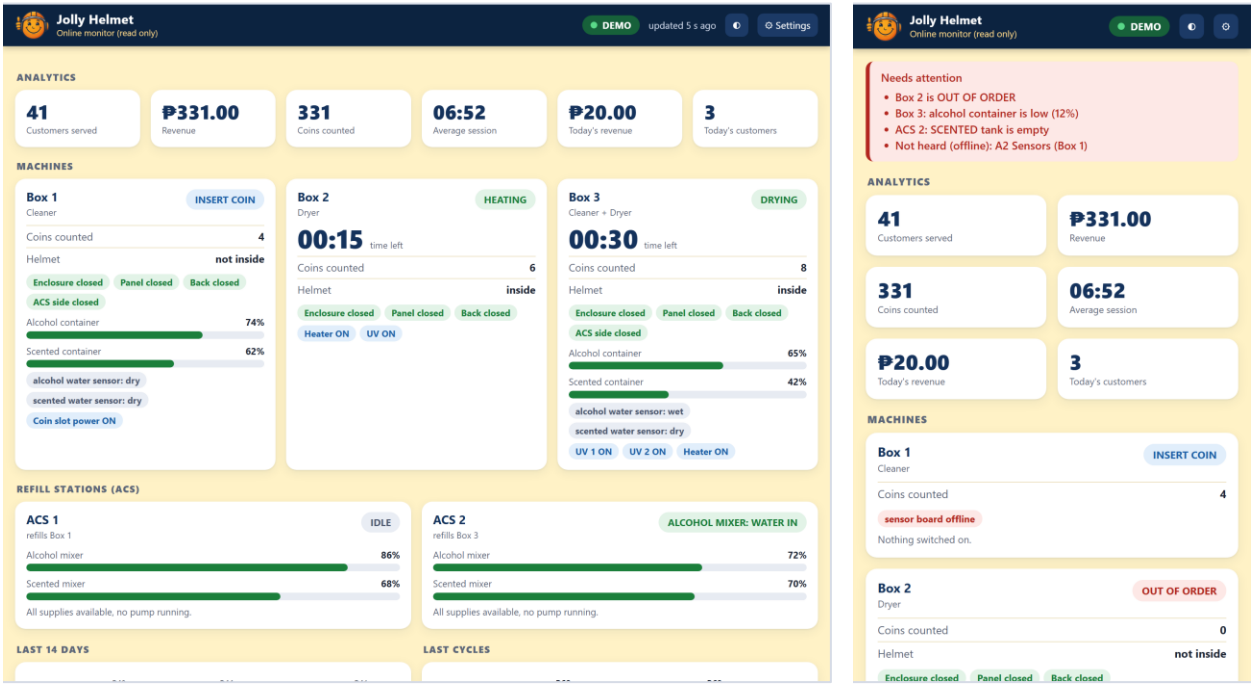
On the setup page	Meaning
ONLINE (green)	The last upload to Blynk worked. It shows how many seconds ago.
NOT SET UP (amber)	No Wi-Fi or no token is saved yet.
TESTING (blue)	The board is joining the Wi-Fi and talking to Blynk.

On the setup page	Meaning
NOT CONNECTED (red)	The Wi-Fi or Blynk failed. The reason is written under the badge.
Machines heard	Must be 13 of 13 when every board is on.
Revenue totals	"heard 4 s ago" means the LocalServer is on. "not heard yet": switch it on or reflash it.
Clock	Set at the first successful upload. The day charts and the secure connection need it.

While you use the setup page the board waits 90 seconds before its next upload, so the page does not drop while you type. A phone left connected to its Wi-Fi is disconnected for a moment at every upload.

Step 3 - Open the web page

1. Put the folder **ONLINE_WEB** on a web host (GitHub Pages, Netlify, Cloudflare Pages, the school server) or open **index.html** on a PC.
2. The first time, the page asks for the Blynk Auth Token. Paste it and click **Connect**. It is kept in that browser only. On a phone you can also open the page once as *index.html#token=THE_TOKEN*.
3. To try the page without hardware, add **?demo** to the address (or **?demo=alerts** to see the warnings).



The web page on a PC and, on the right, on a phone with warnings (demo data)

Reading the web page

Part	What it shows
Badge (top right)	**LIVE** green: fresh data. **DELAYED** amber: a little late. **OLD DATA** red: nothing new for a long time. **BAD TOKEN** or **NO LINK**: the page cannot read Blynk. Next to it: "updated 5 s ago".

Part	What it shows
Needs attention (red box)	A box out of order, a container below 15 %, an empty ACS tank, a low mixer, a board that is not heard.
Analytics	Customers served, revenue, coins, average session, today's revenue and customers.
Machines	The step (green = running, blue = a customer is using it, amber = needs attention, red = out of order), time left, coins, helmet, doors, container levels with a small graph, water sensors and the relays that are ON.
Refill stations	Step, mixer levels, EMPTY tanks, pumps running, warnings.
Last 14 days, Last cycles	Revenue per day (today on the right) and the last 10 sessions (revenue and length).
Boards and Online Server	Online or offline for the 13 boards, the Online Server's uptime, restart reason, Wi-Fi signal, free memory and uploads this month.
Buttons	The half-moon switches light and dark. **Settings**: token, Blynk server and refresh time (5 seconds to 1 minute).

If a box shows "sensor board offline", the page does not hear that board, so it hides that box's doors, helmet and levels instead of showing wrong numbers.

Good to know

- The page only reads. A Blynk token can also write values, so do not publish it: leave it out of the page files and type it in the page.
- Blynk counts messages. One upload a minute is about 43,200 a month, plus one for V1 to V4 and one per alert event. Check your plan's limit.
- The Blynk Console may show the device as offline. This is normal: the board uses Blynk's web API, not a permanent connection. The page decides "alive" from the numbers inside the data.
- The board hears the machines on radio channel 1 and leaves it for about 5 to 10 seconds at every upload, so do not send faster than every 30 seconds.
- Only 2.4 GHz Wi-Fi works (the ESP32 cannot join 5 GHz).

Online troubleshooting

Message	Meaning	What to do
Requested datastream doesn't exist in the product	The template has no V0.	Create V0 (String) in the template and Save it.
Invalid token, BAD TOKEN	The token is wrong or belongs to another device.	Copy it again from Device info.
read Timeout, no answer	Blynk did not answer. The board tries once more with a shorter message.	Check that the Wi-Fi really has internet. Try your regional Blynk server in More settings. The status says if only the SHORT version went through.
network not found, wrong password	The Wi-Fi cannot be joined.	Check the name and password. It must be 2.4 GHz.

Message	Meaning	What to do
the clock could not be set	The Wi-Fi blocks the time servers (NTP).	Allow internet access. Last resort: tick "Skip the TLS certificate check".
Machines heard stays low after an upload	The radio did not return to channel 1.	Switch the Online Server off and on, and tell the builder.
Revenue totals not heard	The LocalServer is off or has the old version.	Switch it on, or reflash it.
V1-V4 failed, event failed	The optional datastreams or the alert event are missing in the template.	Create them, or untick them in the setup page.
The page says OLD DATA	The uploads stopped.	Open the board's setup page and read its status.

Customer status page (a second Blynk device)

The customer page (**status.html**) shows Box 1, Box 2 and Box 3 as FREE, IN USE or OUT OF ORDER. It is public, so it must not use your private token (a Blynk token can also write). It reads a **second Blynk device** that holds nothing but those three boxes.

1. In the Blynk Console open **Devices**, **New Device**, **From template**, and pick the same template as before. Name it *Jolly Helmet Public status*.
2. Open it, tab **Device info**, and copy its **Auth Token**. This token may be public. Never use the private token here.
3. Open the Online Server setup page (Wi-Fi **JollyHelmet-Online**). Paste the token into **Public status device token** and tap **Save and connect**. The row **Public status device** on top then says OK.
4. In the folder ONLINE_WEB open **js/config.js** and write the token in **publicToken**. (If you leave it empty, the QR code carries the token in the address.) Edit the prices and the "Something wrong?" line in the same file.
5. Put the folder on the web host. Open the online page and tap the **QR** button. Paste the public token if it asks. Tap **Print the sign** and stick it on every machine.
6. Test it: scan the QR code with a phone. The boxes must show FREE or IN USE. **status.html?demo** shows made-up data.

The public device holds	Meaning
b	For Box 1, Box 2 and Box 3: the step number, the seconds left and whether the brain is heard. Nothing else.
seq, ts, iv	A counter, the time and the upload interval. The page shows NO INFORMATION when the data is older than about 4 minutes.

If somebody misuses the public token, the worst they can do is write a wrong status into that one device (the next upload overwrites it within a minute) and use up some Blynk messages. They cannot reach the private device, the revenue or the machines. The page costs one more Blynk message per upload.

Online Server firmware of 2026-10-10

Flash ESP32_ONLINESRV again (same settings, see Firmware and flashing). The new version:

- counts every finished session per box, per hour of the day and per weekday, from the brains' own counters, and keeps the counts inside the board (this feeds **Busiest hours**, **Busiest days** and **What each box earns** in the Reports tab);
- writes the public status device when its token is set;
- shows two new rows on its setup page: **Public status device** and **Sessions counted**, and has a button **Reset the per-box and busiest-hours statistics** under *More settings*.

Until it is flashed, the three charts above say "Needs the Online Server firmware of 2026-10-10". Everything else works with the old firmware.

Alerts when the page is closed (Blynk app)

A closed page cannot make a sound. The Blynk app can.

1. On the Online Server setup page tick **Send a Blynk event with code alert**, then **Save and connect**.
2. In the Blynk Console open the template, tab **Events**, and make sure the event with the code **alert** exists (see Step 1). Choose the type **Warning** or **Critical** and switch on its **push notification**. Click **Save**.
3. Install the Blynk app on the phone, sign in with the same account and allow notifications. Every new alert (box out of order, tank empty, container low, board offline) now arrives as a notification, even with the page closed.

The Manual tab and the team password

The customer chapters of this manual are public. The rest (admin, technician, reference) prints the Wi-Fi passwords, so the web page keeps it **encrypted**: the file on the web host is unreadable without the team password, and the page decrypts it in the browser after the password is typed. The team password is not the Blynk token.

After you change this Word file: press Ctrl+A, then F9 in Word (updates the contents and the index) and save. Right-click **MANUAL\build_web_manual.ps1** and choose **Run with PowerShell**. It asks for the team password (use the same one, or a new one and tell the team). It makes a fresh PDF next to the Word file and rewrites the folder **ONLINE_WEB\manual**. Upload that folder again. Word must be installed on that PC.

Online page problems

Symptom	Cause	What to do
No sound when an alert comes	The browser needs one tap. Or the page is closed, or sound is switched off.	Tap **Turn on alert sounds**. Keep the page open. Check Alerts, Sounds (the volume and the On boxes).
The customer page says NO INFORMATION	The public token is missing or wrong, the Online Server is not reflashed, or its data is old.	Check **publicToken** in js/config.js (or the QR address). The setup page row **Public status device** must say OK.

Symptom	Cause	What to do
Busiest hours, days and box earnings are empty	The Online Server has the old firmware, or nothing was counted yet.	Flash the firmware of 2026-10-10 and wait for sessions.
A dial says "watching the trend"	It needs about 10 minutes of readings with the page open.	Wait. The trend is kept in the browser.
The Checklist is empty on another phone	Ticks are saved per browser.	Use **Save a backup** and **Restore a backup**.
The Manual tab shows only the customer guide	The team chapters are locked.	Tap **Unlock** and type the team password.
"That is not the team password"	Wrong password, or the web files were made with another one.	Run build_web_manual.ps1 again and upload the folder.

Maintenance

> Before you open a box: finish or cancel any run, switch the machine off at the mains and unplug it. Wait for the heater to cool.

Schedule

When	What to do
Every day	Home page: every badge ONLINE and no red banner. The online web page says LIVE. Look at the 3 supply tanks in each ACS and the 2 main containers. Wipe the touch screens. Collect the coins and note the revenue.
Every week	Wipe the helmet chamber and the door seals. Open each door during a run: SAFETY PAUSE must show. Listen for each lock clicking. Look for leaks at pump tubes and mixers. Test each UV-C lamp in Maintenance with the door closed (the relay must click; never look at a lit lamp). Check the humidifiers make mist.
Every month	Switch every relay by hand in Maintenance and listen for the click. Clean the ultrasonic sensor faces and the water-level probes. Compare each distance with a ruler (use live). Test the coin slot with every coin. Check screw terminals and plugs are tight and nothing is hot or discoloured. Clean the humidifier discs and the heater and fan air intakes. Test the safety slide of each enclosure door: with the power off, sliding it down must unlock the door by hand.
Every 3 months	Capture the water sensors again, dry and wet, with the real liquids. Check the touch accuracy. With the power off, check the mains wiring, earth and fuses. Take a screenshot of every Control Center settings page and keep it.
Once a year or when worn	Replace the UV-C lamps at the end of their rated hours (their output fades before they stop glowing). Replace pump tubing and slow pumps. Replace any lock that sticks or gets hot. Replace humidifier discs.

Calibrate

What	How (Control Center)
Helmet distance	Box page: put a helmet in, tap **use live**, then Send. Do the same on Box 2 (HEAT).
Container lines	LEVELS: fill to each line, tap **use live**. Bigger number = emptier.
Water sensors	LEVELS: **dry now**, then **wet now** in the real liquid (alcohol conducts poorly).
ACS mixer lines and sensors	ACS page: LEVELS and TIMING, use live and **Work out the thresholds**.
Coin slot	Check the coin pulses on the box page: they must rise by the coin's value. Speed (FAST, MED, SLOW) is set on the acceptor.
Touch screen	Flash ESP32_C1_CALIBRATE on the terminal, touch the 4 targets, copy the calData line into the terminal sketch and flash it again. B1 (Box 3) may still need this.

Test every output by hand

1. Put the box on its start screen (TAPS) or OUT OF ORDER.
2. Control Center, box page, **Maintenance**. Tap each relay and check that the right thing happens.
3. Do the ACS pumps one at a time and compare each with its label before you trust the automatic run.
4. Tap **Maintenance** again. Everything switches off.

Replace a broken board

1. Power off. Photograph the wiring, or print the pin table of that board.
2. Move the wires one at a time to the same GPIO on the new board.
3. Flash the right sketch (see Firmware and flashing). A3 and B3 must be plain ESP32 boards.
4. Power on and wait for the ONLINE badge, then test each output by hand.
5. Brain (S3) or ACS replaced: enter its settings again. LocalServer replaced: the revenue numbers start from zero. Online Server replaced: join its Wi-Fi again and enter the Wi-Fi and the Blynk token; the 14-day history starts again.

Spare parts to keep

- One spare ESP32 per role, flashed and labelled: A1, A2, A3, C1, C2, B1, B2, B3, ACS, plus one ESP32-S3.
- One spare relay board of each size, 2 MC-38 sensors, 2 ultrasonic sensors, 2 water-level sensors.
- One door lock solenoid, one 12 V supply, 2 DC-DC converters, fuses.
- Spare pumps and tubing, one UV-C lamp, one humidifier, a 10 pack of 1N4007 and 10 kΩ, a few 470 to 1000 µF capacitors.

Hardware troubleshooting

Symptom	Likely cause	What to do
Screen goes white when a lock or pump switches	The 5 V dips	Force refresh or Auto hard refresh (Control Center). Add a 470 to 1000 µF capacitor at the TFT supply. Shorter, thicker 5 V wires.
A relay clicks once at power-up	A pin that pulses at boot (GPIO5 on C2)	Harmless. Move it to a boot-safe pin if it bothers you.
A relay stays ON with no command	A boot-active pin, or a wire on the wrong input	Use a boot-safe pin (GPIO13 was moved to 33 on A3 and B3). Check the wiring.
LocalServer restarts when a phone connects	Supply dip (Home page says BROWNOUT)	Own 5 V supply, short cable, a 470 to 1000 µF capacitor.
A board shows OFFLINE	No power, the wrong sketch, or a bench sketch running	Measure 5 V at the board. Flash the right folder. Switch off every bench sketch. All boards use ESP-NOW channel 1.
Door always open or always closed	Magnet too far, or a wrong wire	Check the gap (a few mm) and the wire from S to the GPIO. Closed must read LOW.
Ultrasonic reads 0 or jumps	No echo, dirty face, TRIG and ECHO swapped, loose wire	Clean the face, check both wires, read the board's log.
Water sensor always wet or always dry	Dirty probes, or a bad dry/wet capture	Clean the probes. Capture dry and wet again in the real liquid.
Coins not counted	Coin slot has no power, loose COIN wire, wrong speed setting	The slot is powered only in the pay steps. Measure its 12 V at DC12V then. Check the 10 kΩ and the COIN wire.
A pump runs but the wrong liquid moves	Relay wired to the wrong pump	Run each pump by hand in Maintenance and fix the wiring.
A lock does not release	No 12 V at the solenoid, or the door presses on the latch	Measure the solenoid while its relay is ON. Push the door in a little. The safety slide still opens the enclosure door.
A lock is hot	It was powered too long	Check for a stuck relay. The firmware limits how long a lock stays energized.
A UV-C lamp does not light	No click: relay or wire. Click but no light: fuse, door switch, lamp or its ballast	Test it in Maintenance. Then, with the power OFF and unplugged, check the fuse, the door switch and the lamp.
Helmet stuck after a power cut	Locks are power-to-unlock (the enclosure door also has a safety slide)	Slide the safety lock down (lower left of the enclosure door) at once. When power is back, test the solenoid with Maintenance, ENCLOSURE LOCK.

Reference (Safety, charts, settings and index)

Safety, flowcharts, decision trees, settings, schematics, policies, glossary, versions and index

Safety and emergencies

The customer side of every machine is safe to use. The dangers are behind the panels: 220 V mains, UV-C light, a hot heater and flammable alcohol. Read this page before you open anything.

In an emergency

1. **Stop the run.** Touch **CANCEL**, then **CONFIRM**. The UV-C, the mist and the heater switch off. Box 2 and Box 3 then cool down for a while (Box 2 runs the fan, Box 3 lets everything rest). The coins are **not** refunded.
2. **If a door is forced open during a run**, the machine shows **SAFETY PAUSE** and switches the UV-C, the mist and the heater off at once. It carries on when the door is shut again with the helmet inside.
3. **Fire, smoke, burning smell, sparks or a person in danger:** switch the machine off at the wall socket or the main breaker, step back, and call for help. Do not use water on anything electrical.
4. **A helmet is stuck, or the power is off:** slide the **safety lock down** (lower left of the enclosure door). The door unlocks at once. More in the next block.

> The solenoid locks are **power-to-unlock**: with the power off, or with a broken solenoid, they stay locked. The **enclosure door has a mechanical safety slide** at its lower left. Slide it **down** and the door unlocks, with or without power. Anybody can use it. Never block or remove it.

A helmet is stuck inside

1. **Fastest way: slide the safety lock down.** It is at the lower left of the enclosure door. The door unlocks at once, even with no power or a broken solenoid. If the machine was running it pauses (SAFETY PAUSE) and the UV-C, the mist and the heater switch off. The helmet may be warm.
2. Check that the machine has power. After a power cut, wait until the Control Center Home page shows the boards ONLINE again.
3. Control Center, the box page, **Maintenance**, tap **ENCLOSURE LOCK**. The door unlocks for up to 60 seconds, then rests for 5 seconds. Tap **Maintenance** again when you are done.
4. If the brain of the box stops answering for 15 seconds while a helmet is behind a closed door, the terminal board lets the enclosure lock go by itself.
5. Still locked? Follow the decision tree *The door will not open* in this reference.

Dangers inside the machine

Danger	Where	Rule
220 V mains	UV-C lamps, heaters, power inlet, fuses	Only a qualified electrician works on the mains side. Unplug first and wait before touching anything. The relays switch the live wire, so a clicking relay does not mean the wire is dead.
UV-C light	Box 1: 2 lamps, Box 2: 1, Box 3: 2	UV-C harms eyes and skin. Never look at a lit lamp. Test a lamp only with the door closed. A door switch that cuts the lamp when the door opens is recommended.

Danger	Where	Rule
Hot heater and air	Box 2 and Box 3	Let it cool (the machine runs a cool down). Never cover the air intakes. The heater switched on by hand in Maintenance turns itself off after 5 minutes.
Alcohol	Containers in Box 1 and Box 3, ACS tanks	Flammable. Keep it away from the heater, sparks and flames. Keep the lids closed and wipe spills at once. Keep a fire extinguisher near the machines.
Water and electricity	Pumps, mixers, humidifiers, drain	Fix leaks straight away. Keep tubes and containers away from the boards and relay modules.
Door lock coils	11 solenoid locks, 12 V	A coil held on for a long time overheats (once it caused a fire-risk incident). The machine never holds one on for more than 60 seconds in a row.

Safety features built into the firmware

Feature	What it does
Door during a run	Door opened mid-run: SAFETY PAUSE, UV-C, mist and heater off. The run resumes when the door is closed and the helmet is still inside.
Door shut, no helmet	While paused, if the door is shut with nothing inside, the lock is released so the door can be opened again.
Lock duty cap	A lock is never on for more than 60 seconds in a row, then it rests for 5 seconds.
Brain-loss watchdog	If a terminal hears nothing from its brain for 15 seconds, it switches the heater, UV-C, mist, pumps and coin power off.
Hand-set limits	In Maintenance the heater stops after 5 minutes, locks after 60 seconds and the ACS pumps after 60 seconds.
Cool down	After heating, Box 2 runs the fan only and Box 3 lets everything rest, for 60 seconds by default. Also after a cancel.
Empty containers	With level checks on, a box with too little alcohol or scented liquid shows OUT OF ORDER (Box 3 keeps DRYING) and asks the ACS to refill.
Safety slide	A mechanical slide at the lower left of the enclosure door unlocks it by hand, with no power and no firmware. Opening the door then starts SAFETY PAUSE if a run was on.

First aid

- **Electric shock:** switch the power off first, then help. Do not touch the person while they are still in contact with a live part.
- **UV-C in the eyes or on the skin:** move away from the light, and see a doctor if the eyes hurt or the skin is red.
- **Alcohol or scented liquid in the eyes:** rinse with clean water for 15 minutes and get medical advice.
- **Burn from the heater:** cool it under cool running water and cover it with a clean cloth.

Who to call

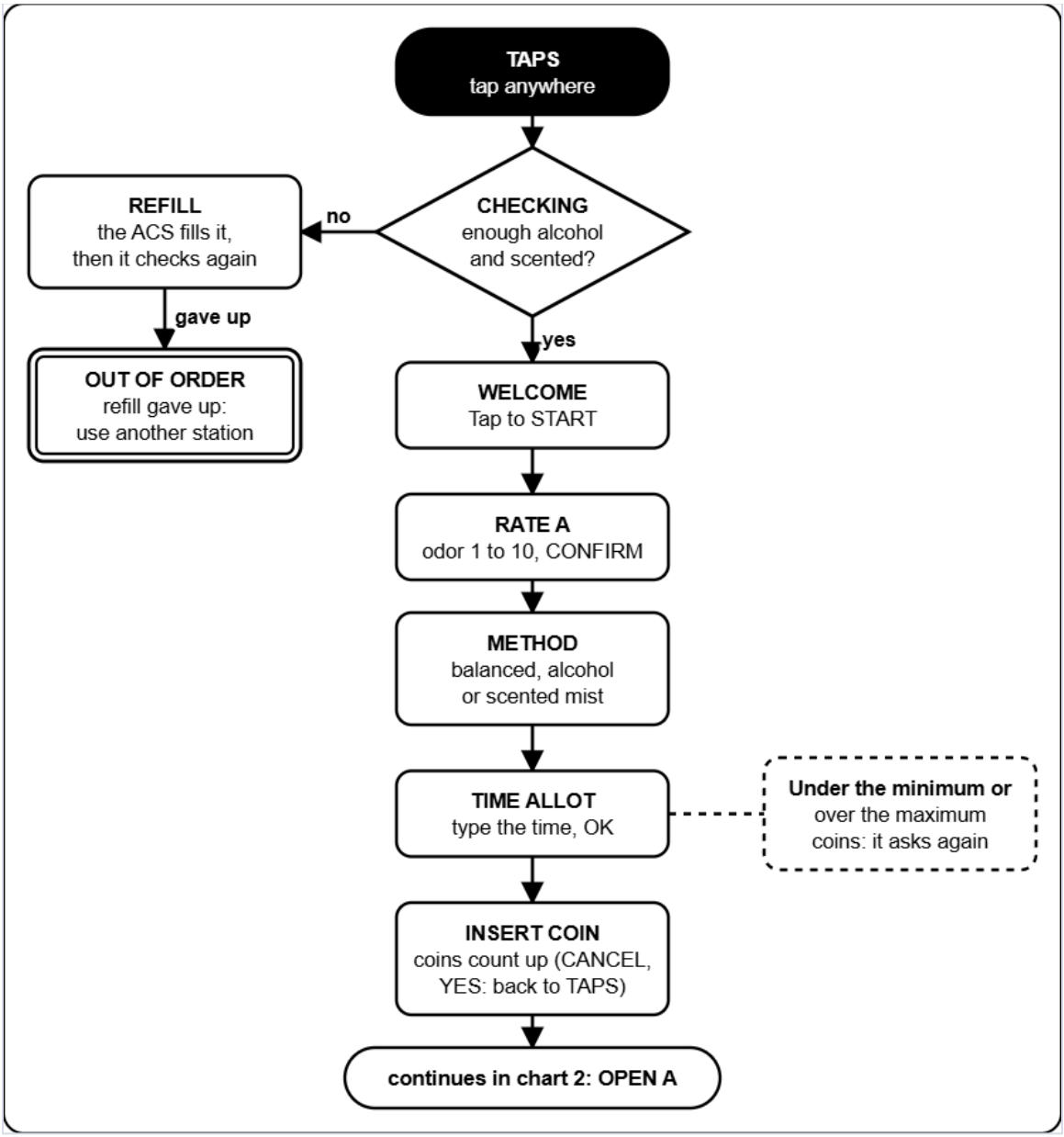
Role	Name	Phone
Owner or supervisor	Sir Jolly Ropal	[Number]
Technician (MEJEZ STUDIO)	Jerahmeel Alzel M Mejes	0975 791 5832
Electrician	Jerahmeel Alzel M Mejes	[0975 791 5832
Fire department	[Local number]	[Number]
Nearest clinic or hospital	[Name]	[Number]

Box flowcharts

One chart per stage of a run, with the same screen names the customer sees. A **black** rounded box is where a chart starts or ends. White boxes are screens or steps. A box with a **thick border** is where the machine is running. A box with a **double border** is a fault. A diamond is a decision. A thin rounded box says where the chart goes on. A dashed box is a note.

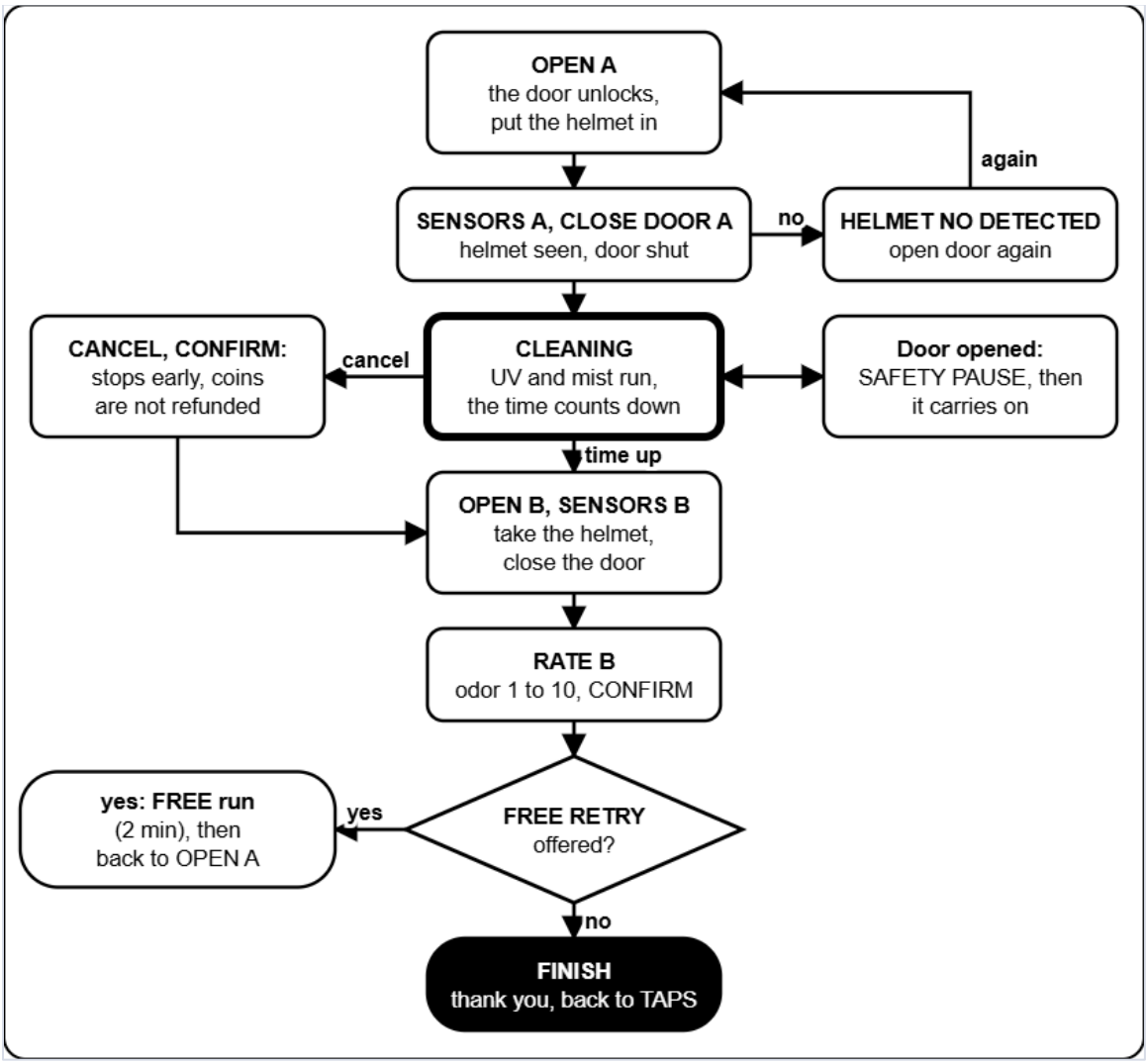
Box	Charts
Box 1 Cleaner	1. From TAPS to INSERT COIN. 2. Open A, cleaning, open B, rating, free retry, finish.
Box 2 Dryer	3. From TAPS to INSERT COIN and door A. 4. Heating, cool down, open B, rating, free retry, finish.
Box 3 Cleaner + Dryer	5. From TAPS to INSERT COIN, with SELECT MODE. 6. Cleaning or drying, cool down, open B, rating, free retry, finish.

Box 1, chart 1: paying



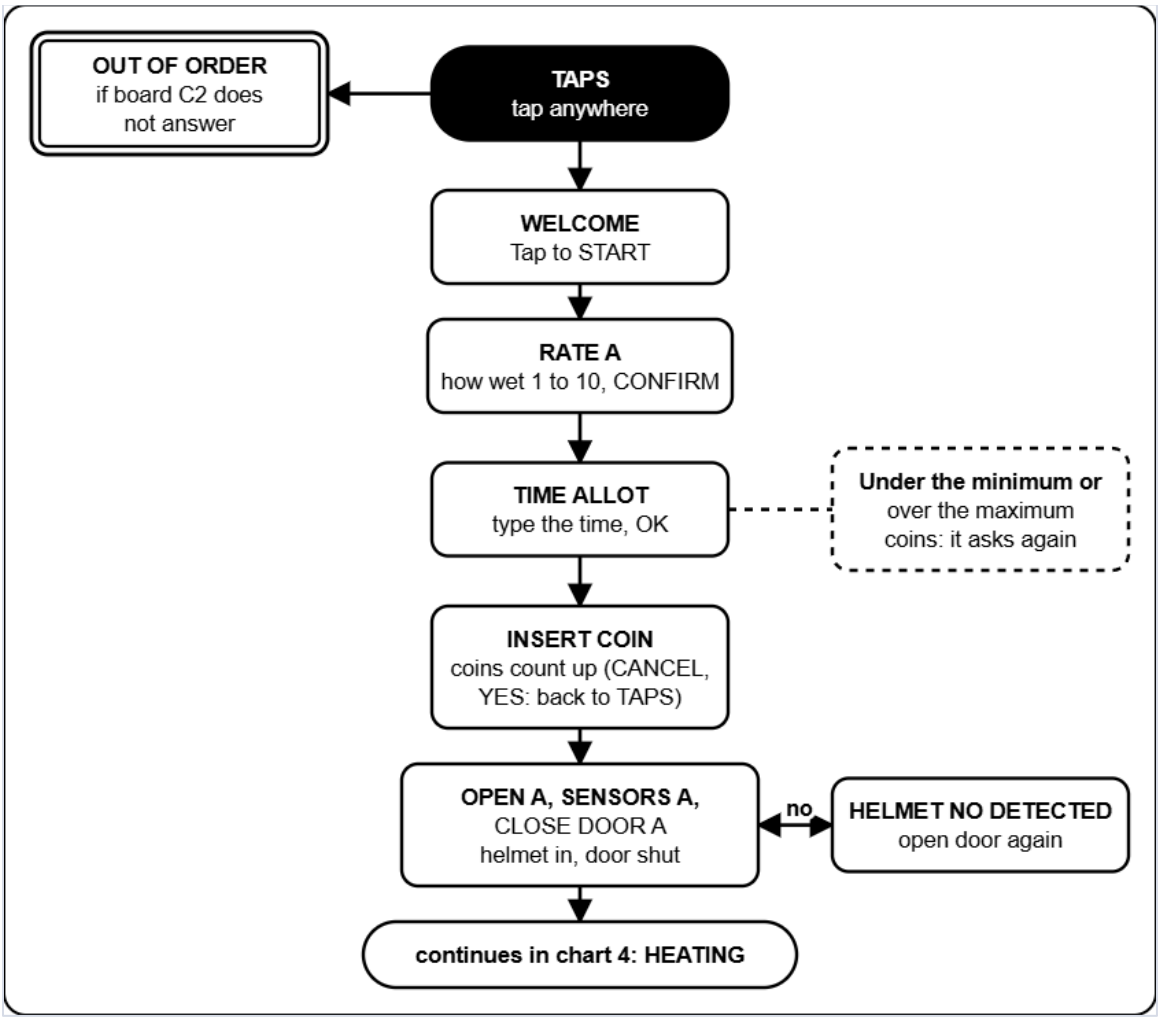
Box 1: from TAPS to INSERT COIN. Containers are checked first; the ACS refills them if they are low.

Box 1, chart 2: the run



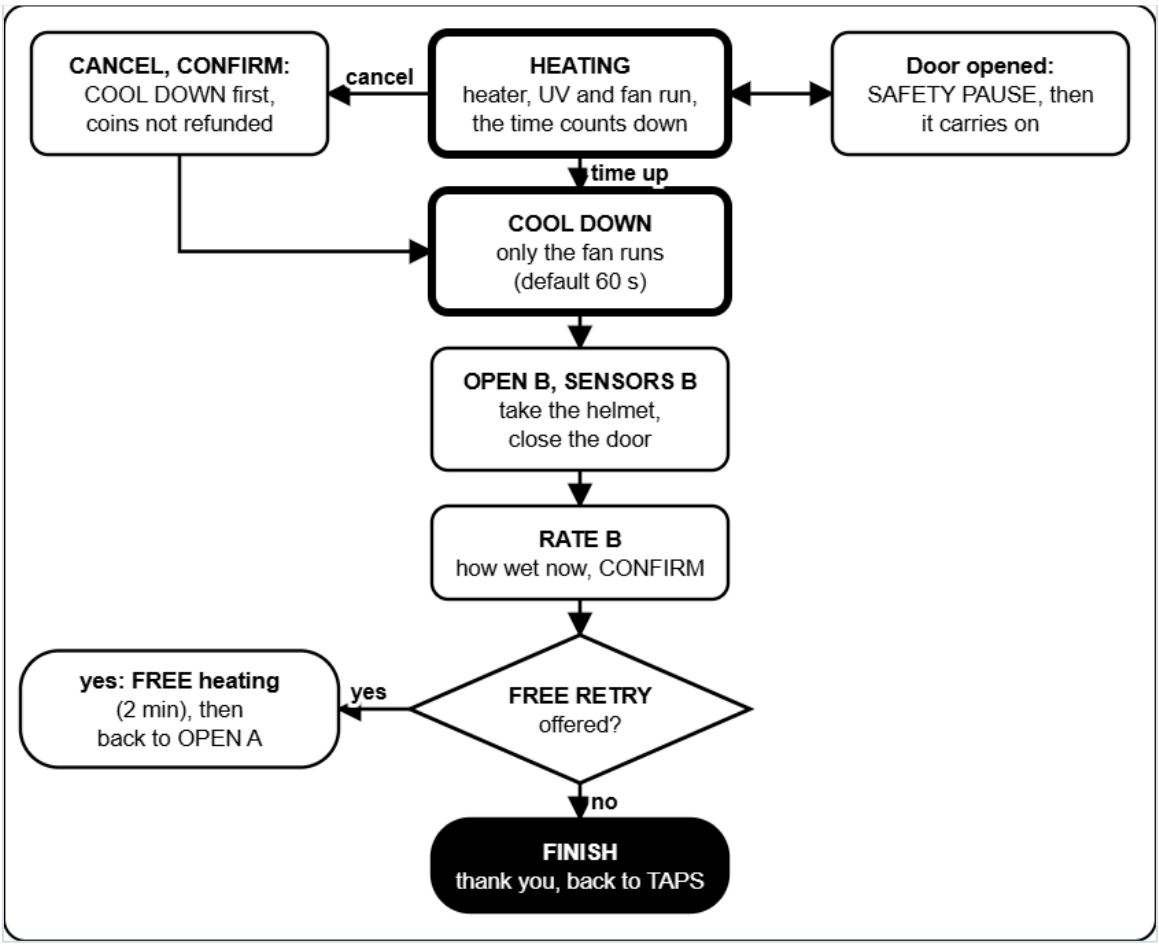
Box 1: cleaning, taking the helmet out, the second rating and the free retry.

Box 2, chart 3: paying



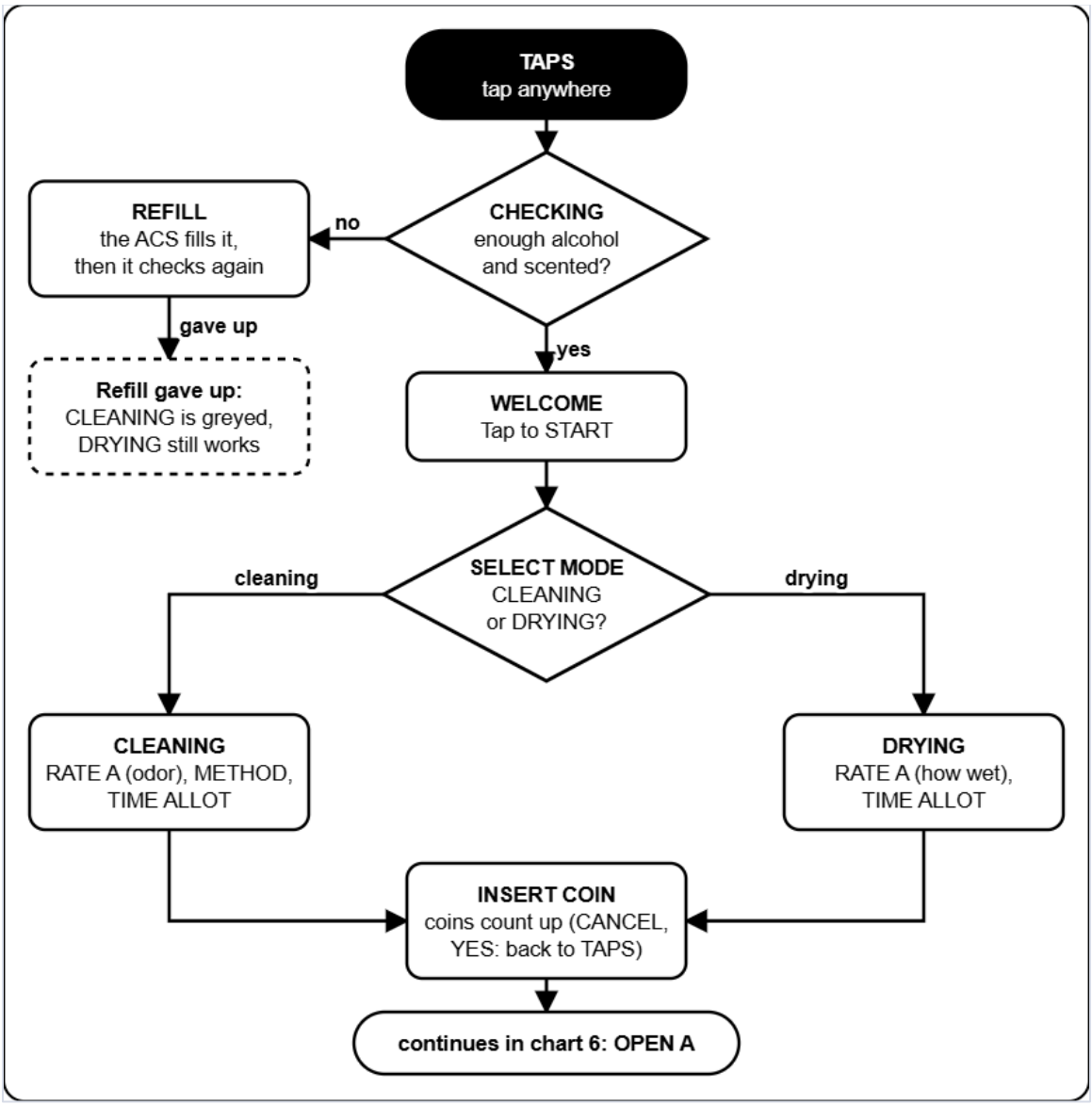
Box 2: from TAPS to the door. The box has no containers to check; it is out of order only if its terminal board stops answering.

Box 2, chart 4: the run



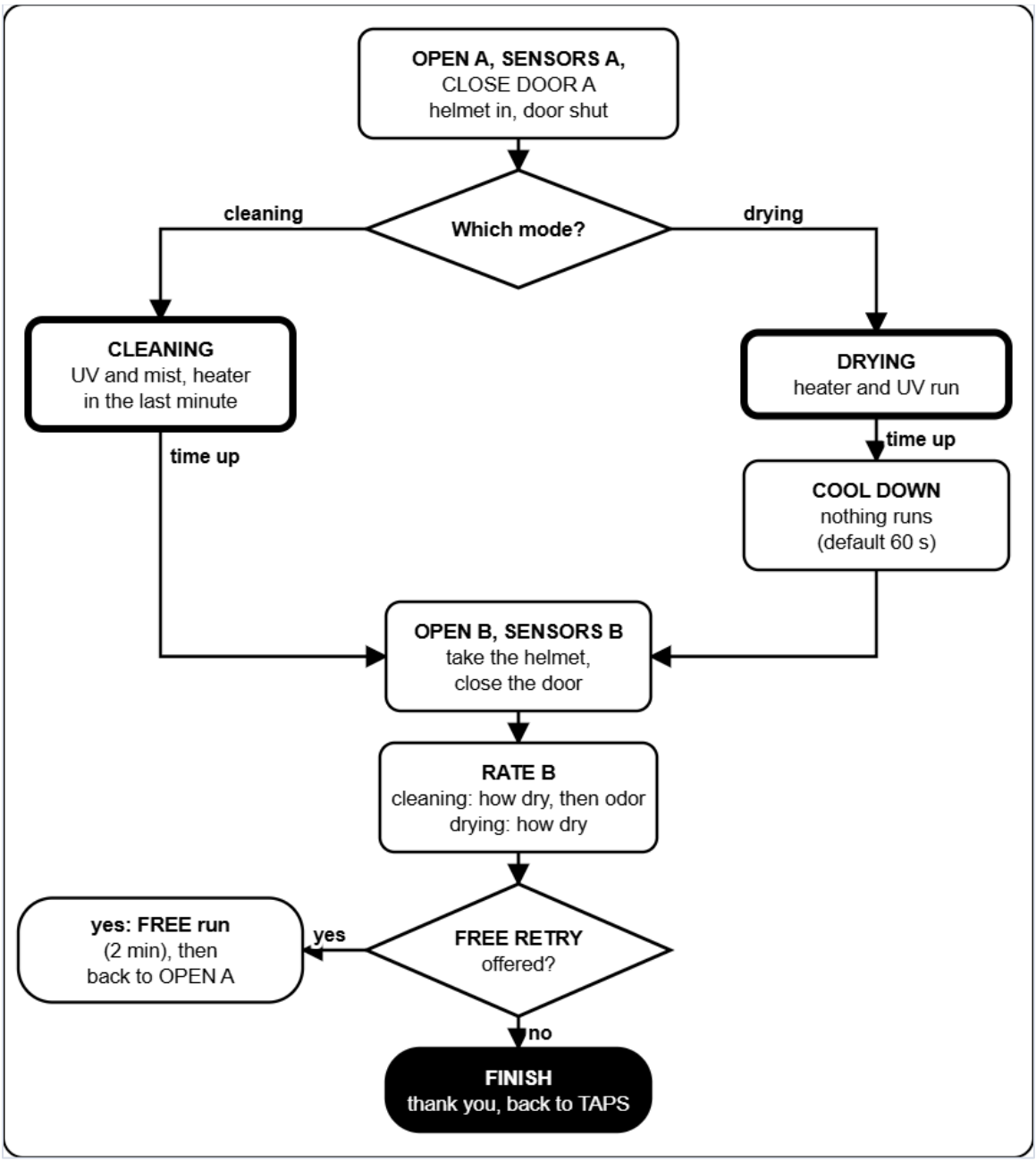
Box 2: heating, cool down with the fan, taking the helmet out and the free retry.

Box 3, chart 5: paying



Box 3: the customer chooses CLEANING or DRYING. If the refill gives up, CLEANING is greyed out and DRYING still works.

Box 3, chart 6: the run



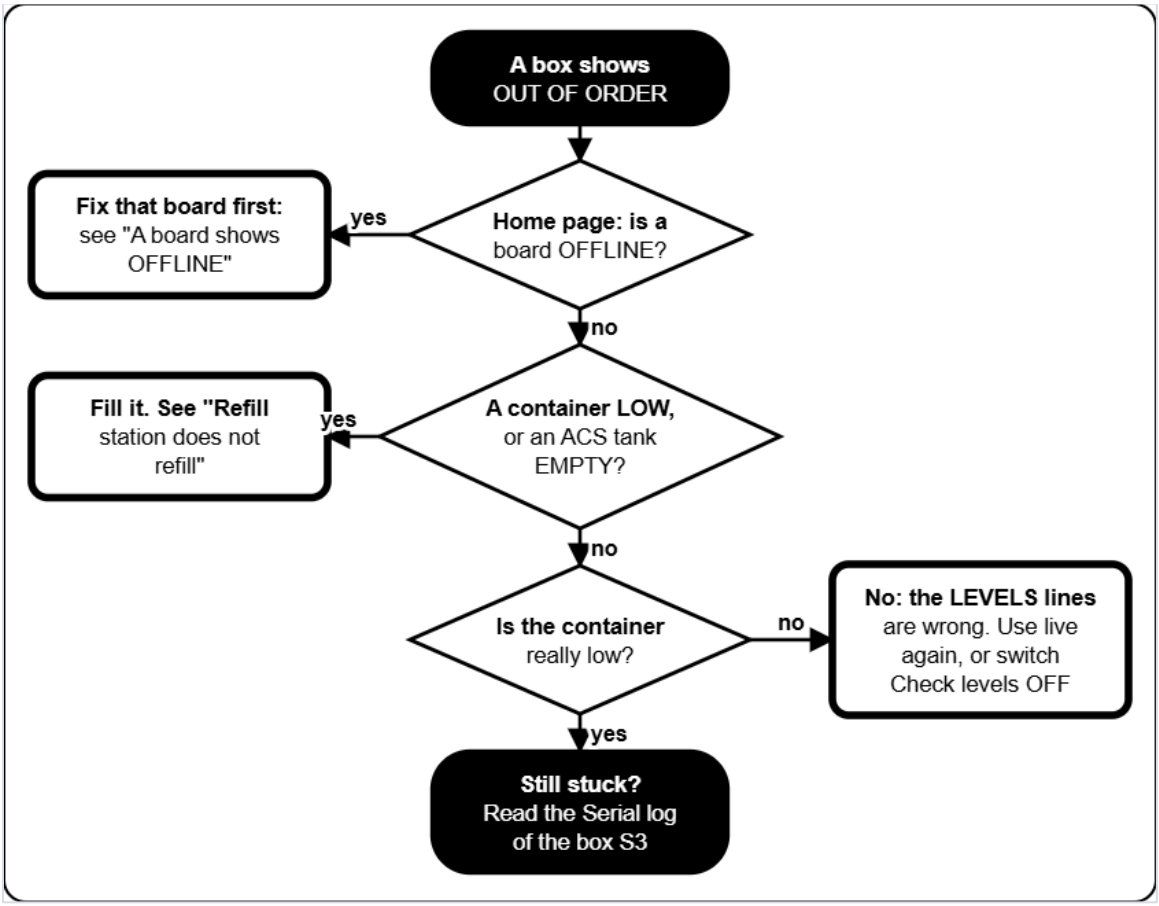
Box 3: cleaning (UV-C and mist, heater in the last minute) or drying (heater and UV-C), then cool down, the second rating and the free retry.

Troubleshooting decision trees

Start at the black box at the top. Answer each question (the diamonds) with yes or no and follow the arrow. Boxes with a thick border tell you what to do. These trees cover the problems that happen most; the tables in the Admin and Technician guides have the rest.

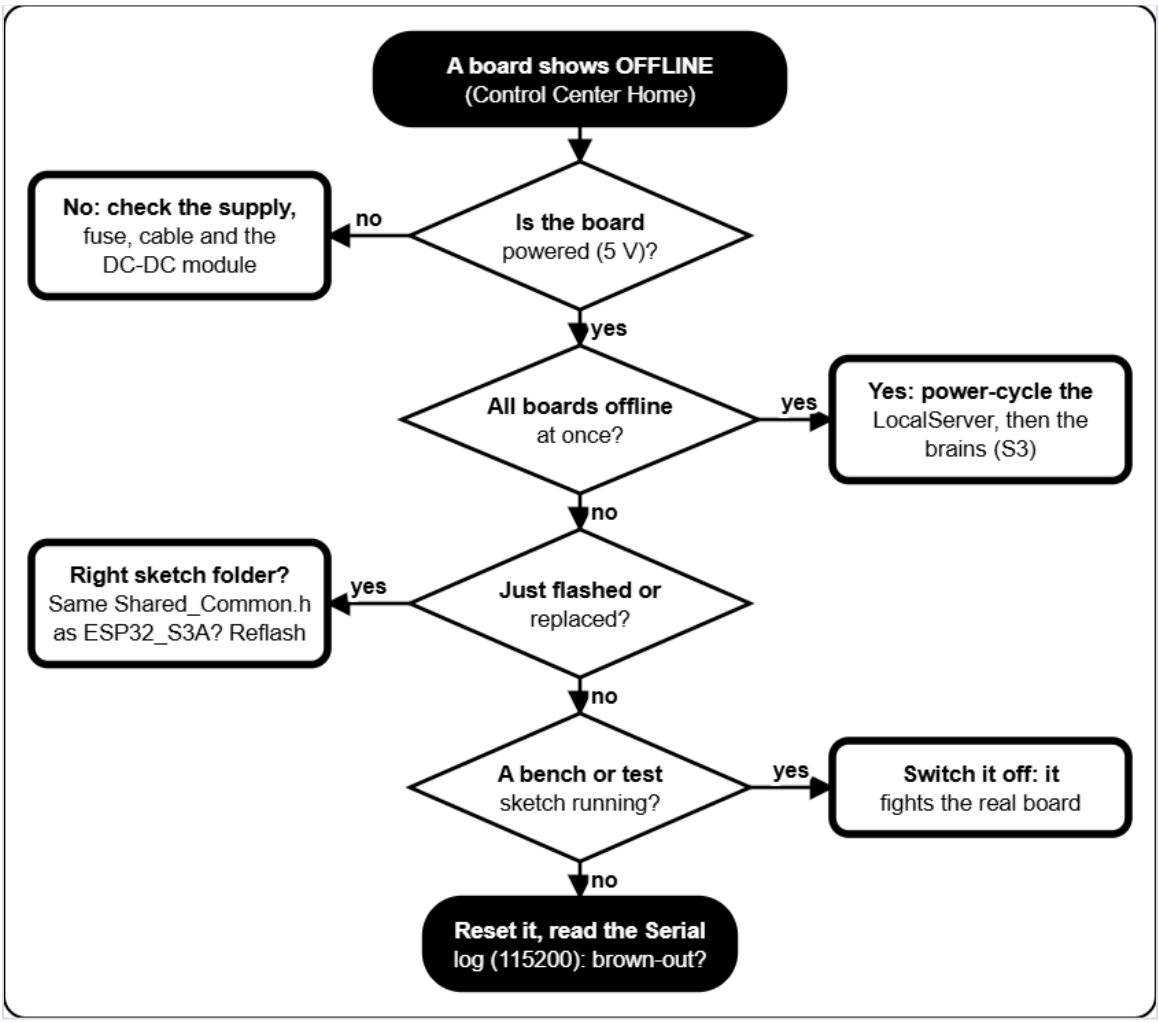
Symptom	Tree
The screen says OUT OF ORDER	1. A box shows OUT OF ORDER
A board is OFFLINE on the Home page	2. A board shows OFFLINE
A coin goes in and nothing counts	3. Coin inserted, the count does not move
A helmet is trapped, a door will not open	4. The door will not open
The refill station does not fill	5. A mixer is LOW or a tank is EMPTY
The web page shows old data or no link	6. The web page says OLD DATA or NO LINK
The touch screen is white, blank or wrong	7. Screen white, blank or the touch is wrong

Tree 1: a box shows OUT OF ORDER



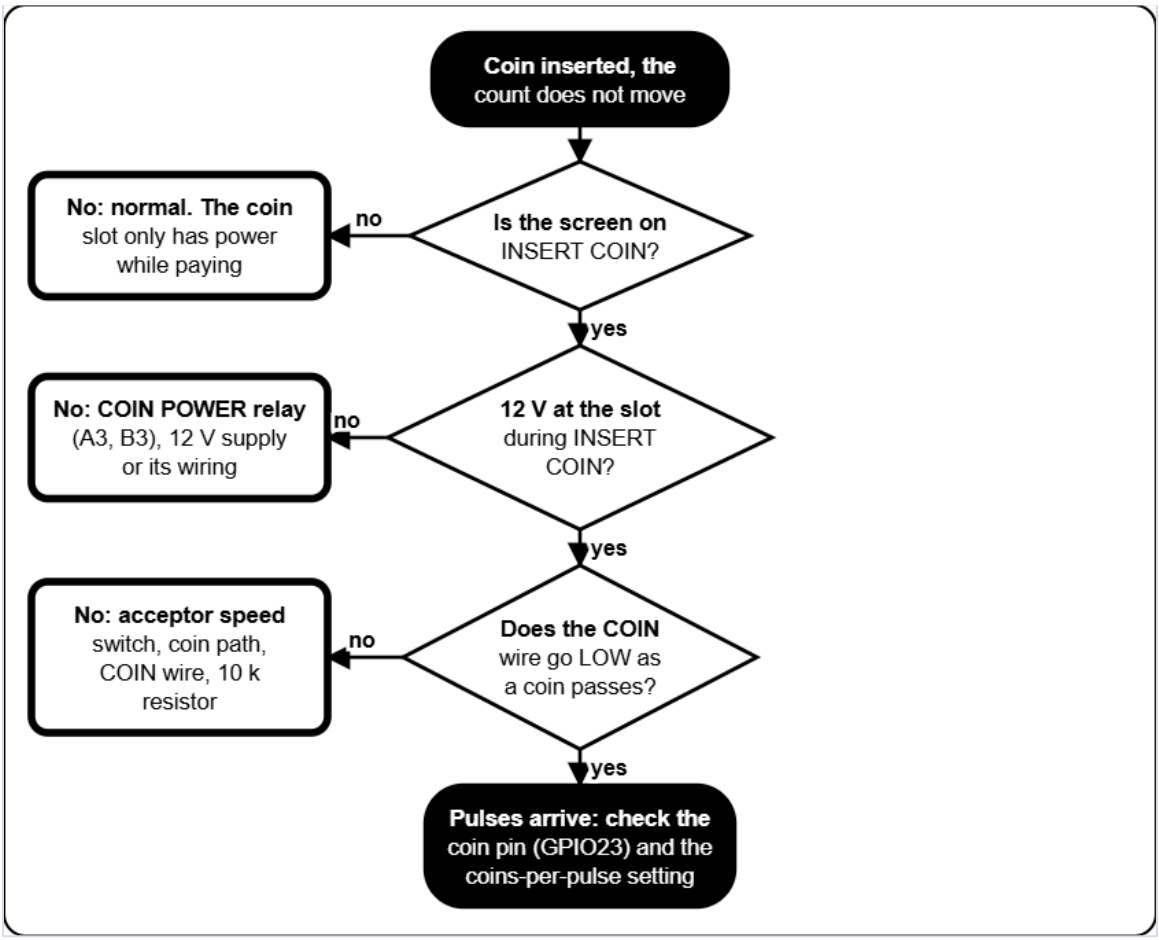
OUT OF ORDER means a container is too low and the refill gave up, or a board is missing.

Tree 2: a board shows OFFLINE



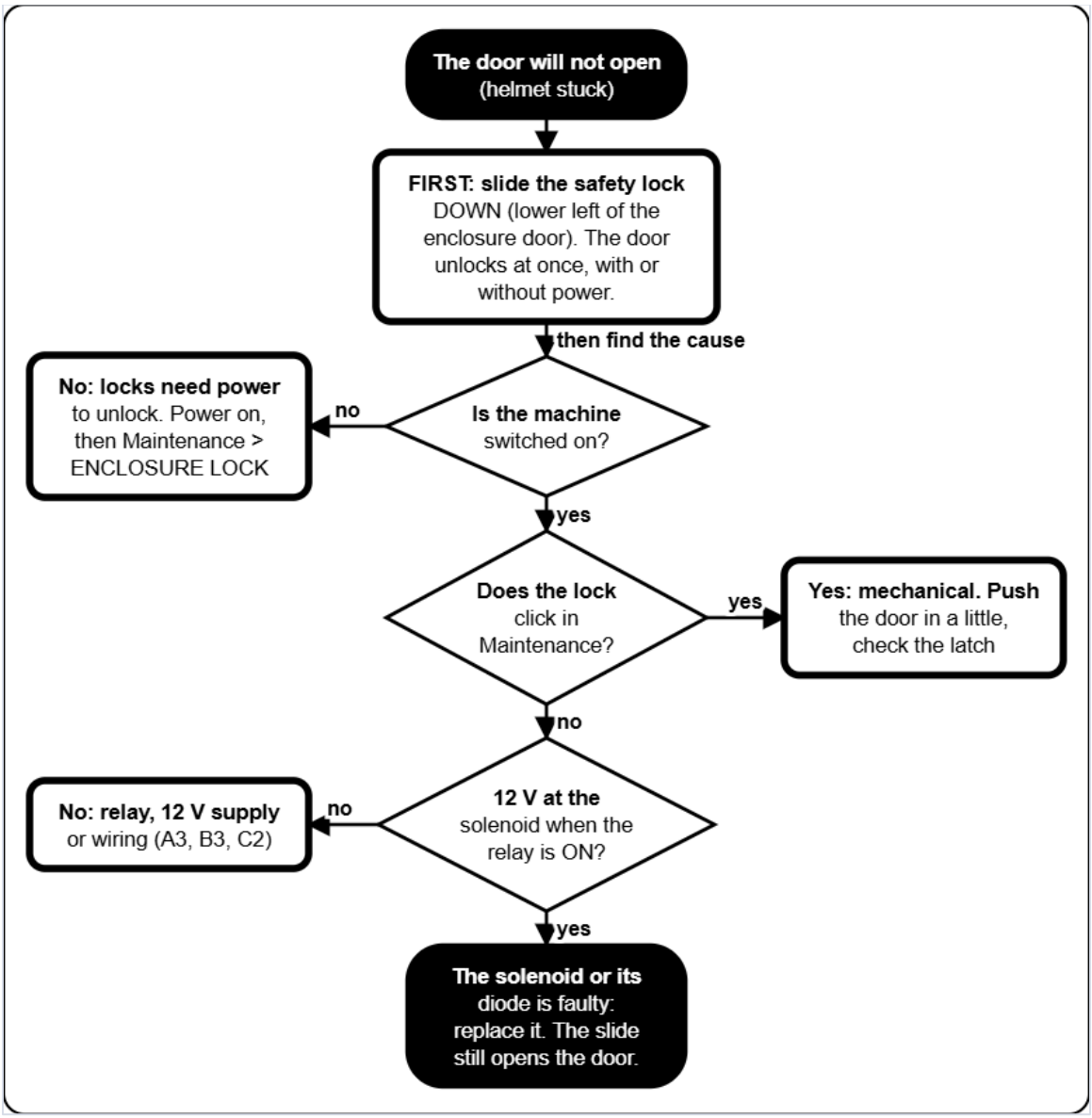
A board is OFFLINE when the LocalServer has heard nothing from it for a while.

Tree 3: coin inserted, nothing counts



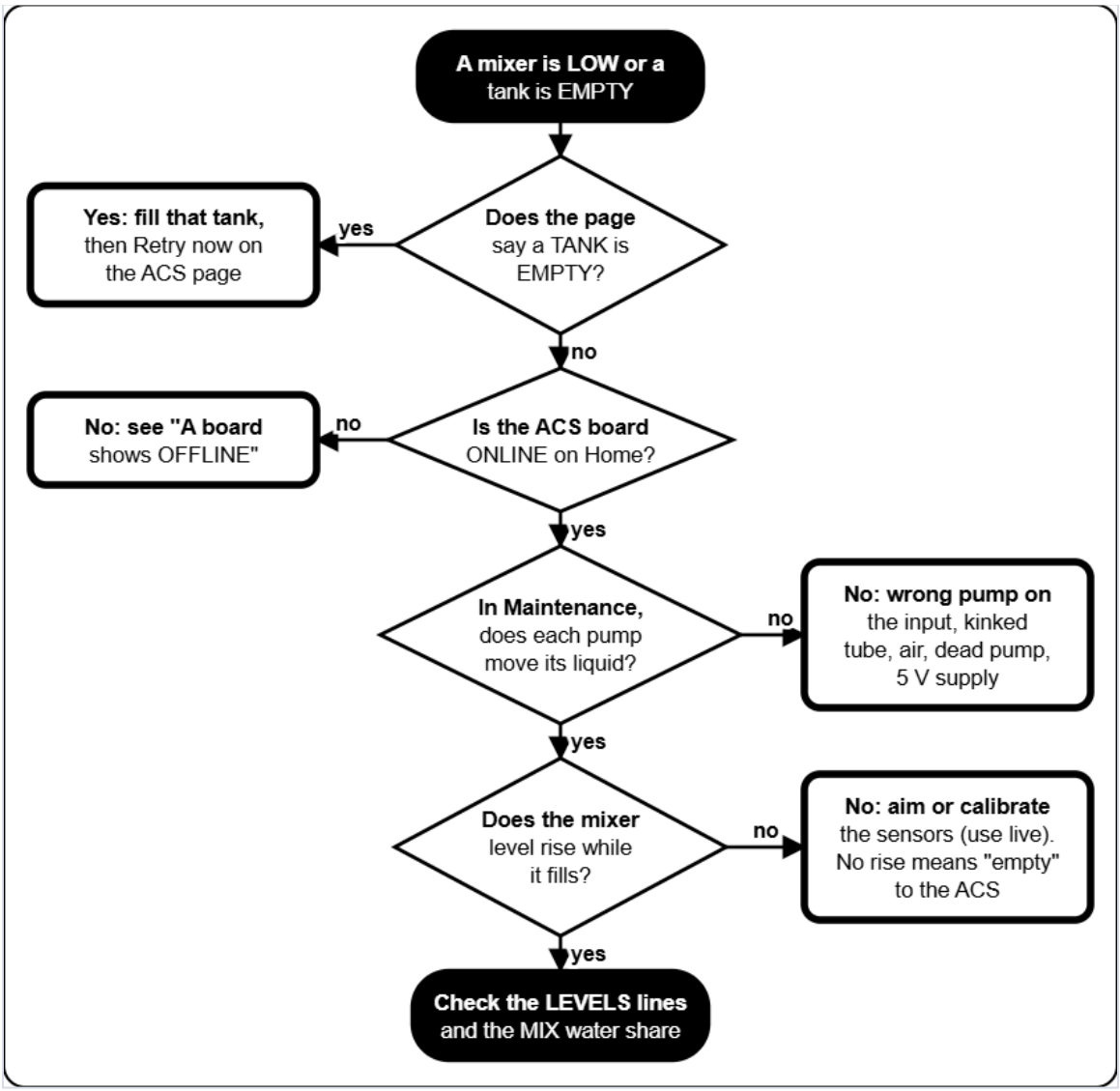
In Box 1 and Box 3 the coin slot has power only while the screen shows INSERT COIN.

Tree 4: the door will not open



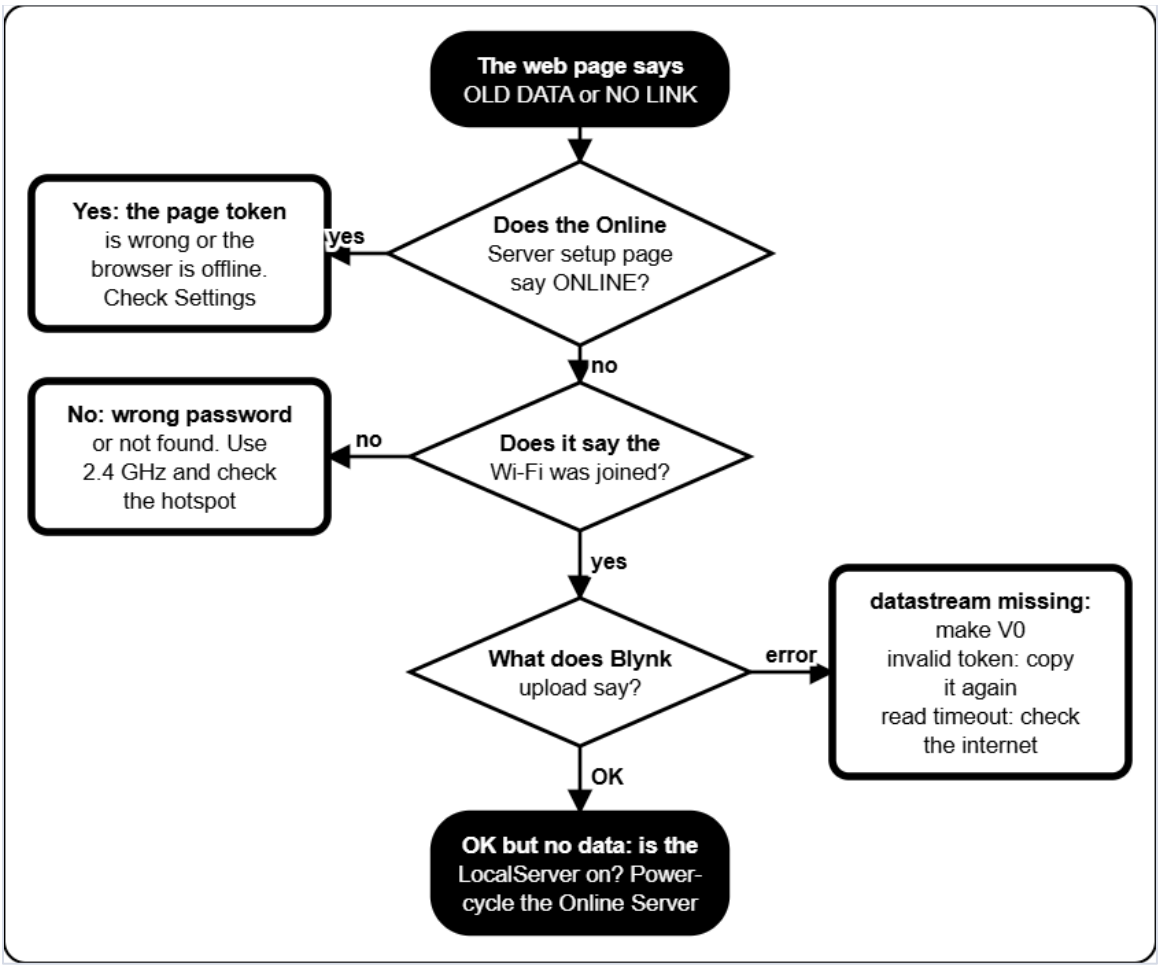
The locks are power-to-unlock: with no power a solenoid door stays shut, but the safety slide (lower left of the enclosure door) opens it by hand. Switch the mains off before reaching in.

Tree 5: the refill station does not fill



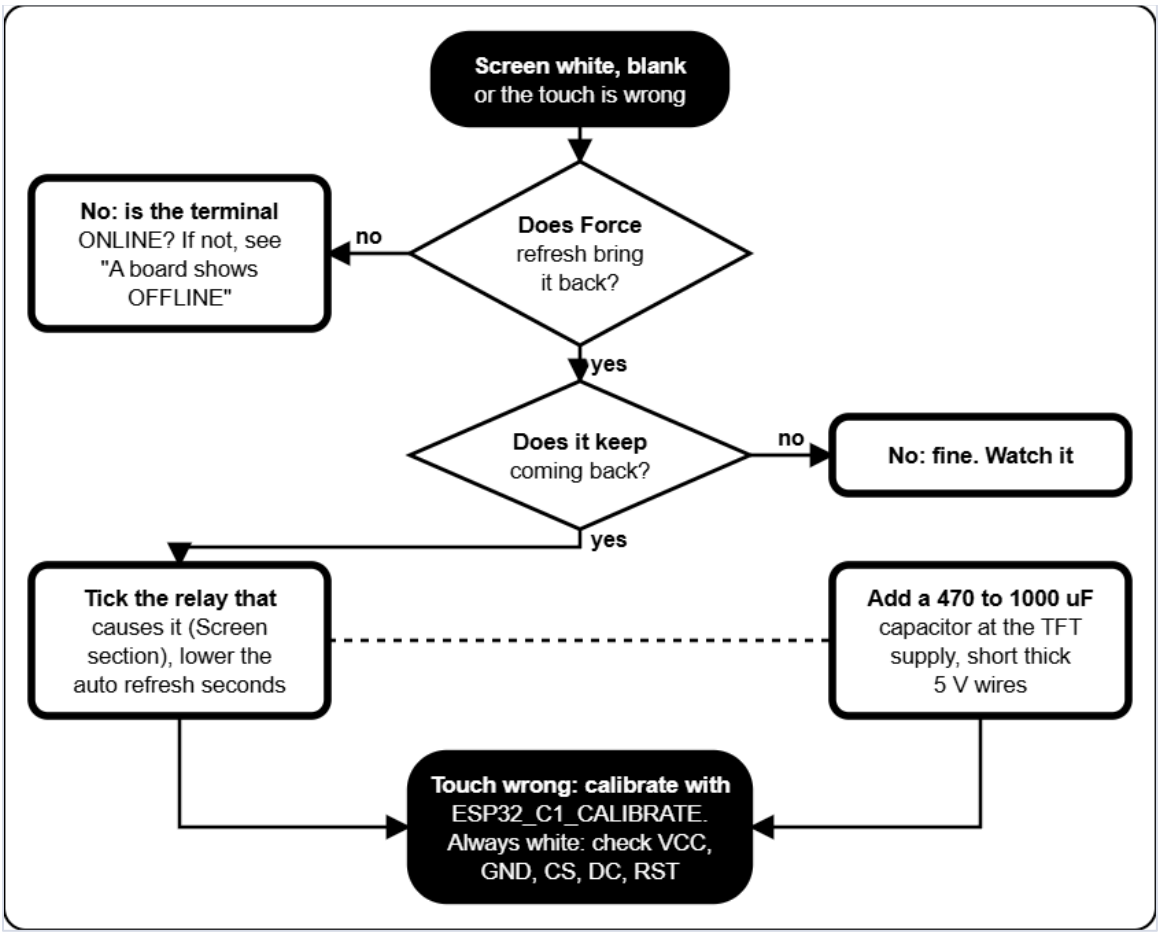
The ACS reads "level did not rise" as "the supply tank is empty".

Tree 6: the web page shows old data



The web page is read only. It shows what the Online Server last sent to Blynk.

Tree 7: the touch screen is white or wrong



A white screen is almost always a supply dip when a heavy load switches. Force refresh re-initialises the screen.

Settings reference

Every setting you can change, with the value the firmware starts with and the range it accepts (a value outside the range is refused). Write the value you use in the last column. The firmware defaults are what a freshly flashed board starts with; the machine you are looking at may have been changed. Settings are saved inside the boards, so they survive a power cut and a LocalServer swap. Where to change them: Control Center, the box page (or **/acs**).

Prices and time (Box 1, Box 2 and Box 3)

Setting	What it does	Default	Allowed	Yours
Rating bands (4)	From and to rating, and the minutes suggested for that rating.	Box 1, 3: 1-5 = 5, 6-7 = 10, 8-9 = 15, 10 = 20 min. Box 2: 1-3 = 3, 4-6 = 5, 7-8 = 8, 9-10 = 10 min	Rating 1-10 (from not above to), 1-120 min	
Minimum time	Shortest session the customer can buy.	Box 1, 3: 5 min. Box 2: 3 min	1-120 min	
Coin rate	N coins buy X minutes (or seconds).	Box 1, 3: 15 coins = 5 min (20 s a coin). Box 2: 10 coins = 3 min (18 s a coin)	1-500 coins, 0.1-9999 time	
Most coins	Largest number of coins one session can take.	Box 1, 3: 90. Box 2: 100	1-999	
Free retry	One free run after the final rating.	On	On or off	
Free retry from rating	Lowest final rating that gets the free retry (1 = everyone).	1	1-10	
Free retry length	How long the free run lasts.	2 min	1-60 min	
Cool down (Box 2, Box 3)	How long the cool down lasts, also after a cancel.	60 s	5-900 s	

Containers and sensors (Box 1 and Box 3)

The distances are what the ultrasonic sensor on the main container reads, from the sensor down to the liquid. A **bigger** number means **emptier**. The defaults are placeholders: measure your real containers.

Setting	What it does	Default	Allowed	Yours
Check levels	Refuse to run (OUT OF ORDER) when a container is low.	Off	On or off	

Setting	What it does	Default	Allowed	Yours
Alcohol low	At or over this distance the alcohol container is LOW.	20 cm	0-400 cm	
Alcohol full	At or under this distance it counts as refilled.	8 cm	0-400 cm	
Scented low	Same for the scented container.	20 cm	0-400 cm	
Scented full	Same for the scented container.	8 cm	0-400 cm	
Helmet detect	The helmet counts as inside when its sensor reads under this.	Box 1, 3: 15 cm. Box 2: 30 cm	1-200 cm	
Water sensor WET (alcohol, scented)	Raw reading at which the sensor means wet.	600	0-4095	
Water sensor DRY (alcohol, scented)	Raw reading at which it means dry again (the gap stops flicker).	400	0-4095	
Reading goes up when wet	Direction of the water sensor.	Yes	Yes or no	
Refill tries	Refill attempts before the box gives up.	10	1-99	
Refill interval	Seconds between tries.	10 s	1-250 s	
Ask the ACS to refill	Off: only count the tries. On: really send the refill request.	Off	On or off	

Check levels and Ask the ACS to refill are **off** when a board is flashed, so a box with uncalibrated sensors still serves customers. Measure the four distances, then switch both on.

Cleaning phases (Box 1 and Box 3)

A paid run is cut into a start part, a middle part (all that is left) and an end part. Each relay is on or off in each part. The lengths are in percent, seconds or minutes.

Setting	What it does	Default	Allowed	Yours
Phases in a paid run	Use the start, middle and end parts.	On	On or off	
Phases in the free retry	Same, for the free run.	Box 1: off. Box 3: on	On or off	
Start part, end part	Length of the first and the last part.	1 min each	0-7200 (0-100 if percent)	
Humidifiers (3)	Parts in which each mist maker runs.	Middle only (Box 3 free: start only)	Any of start, middle, end	
UV-C lamps (2)	Parts in which each lamp is on.	Start and end (Box 3 free: all)	Any of start, middle, end	

Setting	What it does	Default	Allowed	Yours
Heater (Box 3 only)	Parts in which the heater runs while cleaning.	End (also in the free retry)	Any of start, middle, end	

Screen refresh (all boxes)

Setting	What it does	Default	Allowed	Yours
Refresh when a relay switches on or off	Which relay changes force a full re-initialisation of the touch screen (the white-screen cure). One tick per relay, for ON and for OFF.	Box 1, 2: when the enclosure lock turns off. Box 3: heater on; lock and heater off	Any relay	
Auto refresh	Every N seconds the box re-sends the current page as a full re-init. Screens where the customer types or picks a rating are skipped.	2 s	0 (never) to 60 s	

Refill station (ACS 1 and ACS 2)

The ACS keeps two mixer containers (alcohol and scented) topped up from three supply tanks (water, alcohol, scented). Name in brackets: the name used on the /acs page and the serial console. A, S = the alcohol and the scented mixer. Distances are from the ultrasonic sensor down to the liquid.

Setting	What it does	Default	Allowed	Yours
Empty (emptyA, emptyS)	Sensor to the bottom of the mixer, for the level percent.	14 cm	5-300 cm	
Top-up at (topupA, topupS)	Distance at which a top-up starts.	8 cm	1-300 cm	
Low (lowA, lowS)	At or over this distance the mixer is LOW.	11 cm	1-300 cm	
Full (fullA, fullS)	At or under this distance the mixer is FULL.	3.3 cm	1-300 cm	
Water share (waterA, waterS)	Water part of each top-up.	Alcohol 30 %, scented 90 %	1-99 %	
No rise (norise)	A pump runs this long and the level does not rise: that supply is called empty.	12 s	3-120 s	
Rise (risecm)	How far the distance must shrink to count as a rise.	0.5 cm	0.1-5 cm	
Water rise (wlrise)	How far the water-sensor reading must move to count as a rise.	150	20-4000	
Longest fill (phasemax)	Hard limit for one fill pump run.	120 s	10-900 s	
Delivery shot (deliver)	One delivery to the box.	6 s	1-60 s	

Setting	What it does	Default	Allowed	Yours
Try again after (emptyhold)	An empty supply is tried again after this.	300 s	10-3600 s	
Request valid for (reqhold)	A request from a box stays valid this long.	30 s	5-300 s	
Pause after a time limit (holdoff)	Fills pause this long after a pump hit its time limit.	60 s	5-900 s	
Water WET, DRY (wlenter, wlleave)	Raw readings for wet and dry on the mixer water sensor.	600, 400	0-4095	
Reading goes up when wet (wlhigher)	Direction of that sensor.	Yes	Yes or no	

LocalServer and Online Server

Setting	What it does	Default	Allowed	Yours
Control Center Wi-Fi	Name and password of the LocalServer Wi-Fi. Open 192.168.4.1.	ESP32_LOCAL, 12345678	Set in the firmware	
Coin value	Pesos one coin pulse adds to the revenue.	1.00	Any positive number	
Online Server setup Wi-Fi	Wi-Fi to open the setup page on.	JollyHelmet-Online, 12345678	Set in the firmware	
Send to Blynk every	How often the Online Server uploads.	1 minute	30 s, 1, 2 or 5 min	
Time zone	Hours from UTC, for the daily revenue.	As set in the portal	-12 to 14	
Extra datastreams, events, skip TLS check	V1 to V4, the alert event, and the certificate check.	Off, off, off	On or off	
Blynk server	Server name.	blynk.cloud	Text	
Public status device token	Token of a second Blynk device for the customer page (status.html). Type - to remove it.	Empty (not used)	A token	
Reset statistics	Button in More settings: starts the per-box, busiest-hours and busiest-days counts again from zero.	-	Button	

Online web page (kept in each browser)

Setting	What it does	Default	Allowed	Yours
Refresh every (Settings window)	How often the page reads Blynk.	10 s	5, 10, 30, 60 s	

Setting	What it does	Default	Allowed	Yours
Early warning for containers (Alerts tab)	The dial turns amber and a notice alert starts here.	30 %	Off, 20-40 %	
Play alert sounds, **Volume**	Master switch and loudness on this device.	On, 80 %	On or off, 0-100 %	
Repeat a critical alert every	An unacknowledged critical alert sounds again.	1 minute	Never, 30 s, 1, 2, 5 min	
Sound of each alert	Which sound, or none, each kind of alert plays.	See Alerts and sounds	9 sounds or silent	
Chime when a warning clears	Plays when an alert goes away.	On	On or off	
Browser notifications	A notification when the page is in the background.	Off	On or off	
publicToken (js/config.js)	Token of the public status device for the customer page.	Empty (the QR address carries it)	A token	
prices, contact (js/config.js)	Texts on the customer page.	The firmware prices	Any text	
Team password	Opens the encrypted chapters of the Manual tab.	Set when build_web_manual.ps1 is run	6 or more characters	

Fixed values (not settings)

Value	Where it comes from
Radio channel 1	ESP-NOW and the Wi-Fi of the LocalServer must stay on channel 1.
15 s watchdog	No sign of the brain for 15 s: relays off, a trapped helmet is released.
60 s lock, 5 s rest	A lock never stays on longer than that.
5 min heater, 60 s pumps	Limit of a hand-set heater, and of a hand-set ACS pump, in Maintenance.
One free retry	A session can have only one.
Coin signal on GPIO 23	On A1, B1 and C1.
Safety slide	Mechanical, lower left of the enclosure door. Not a setting: it always works.

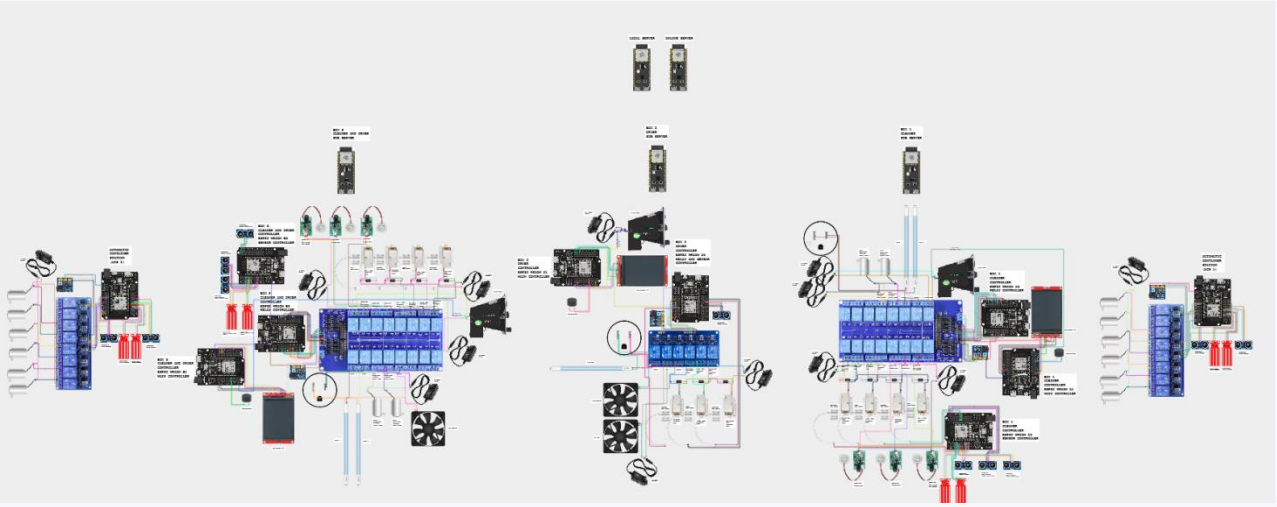
Circuit schematics

The wiring of the machines was drawn in Cirkit Designer. The pictures below are exports of that drawing. Where a picture is missing a dashed box shows the file name to use: save the export in **MANUAL\images\cirkit** under that name, then rebuild the manual (see the README). The drawings in the Technician guide (power, relay stage, locks, coin slot, Online Server) are a simplified view of the same wiring.

Picture	Shows	File name
Whole installation	All boards and how they are powered	cirkit_overview
Box 1	A1, A2, A3, S3A	cirkit_box1
Box 2	C1, C2, S3C	cirkit_box2
Box 3	B1, B2, B3, S3B	cirkit_box3
Refill station	ACS 1 and ACS 2: pumps, mixers, sensors	cirkit_acs

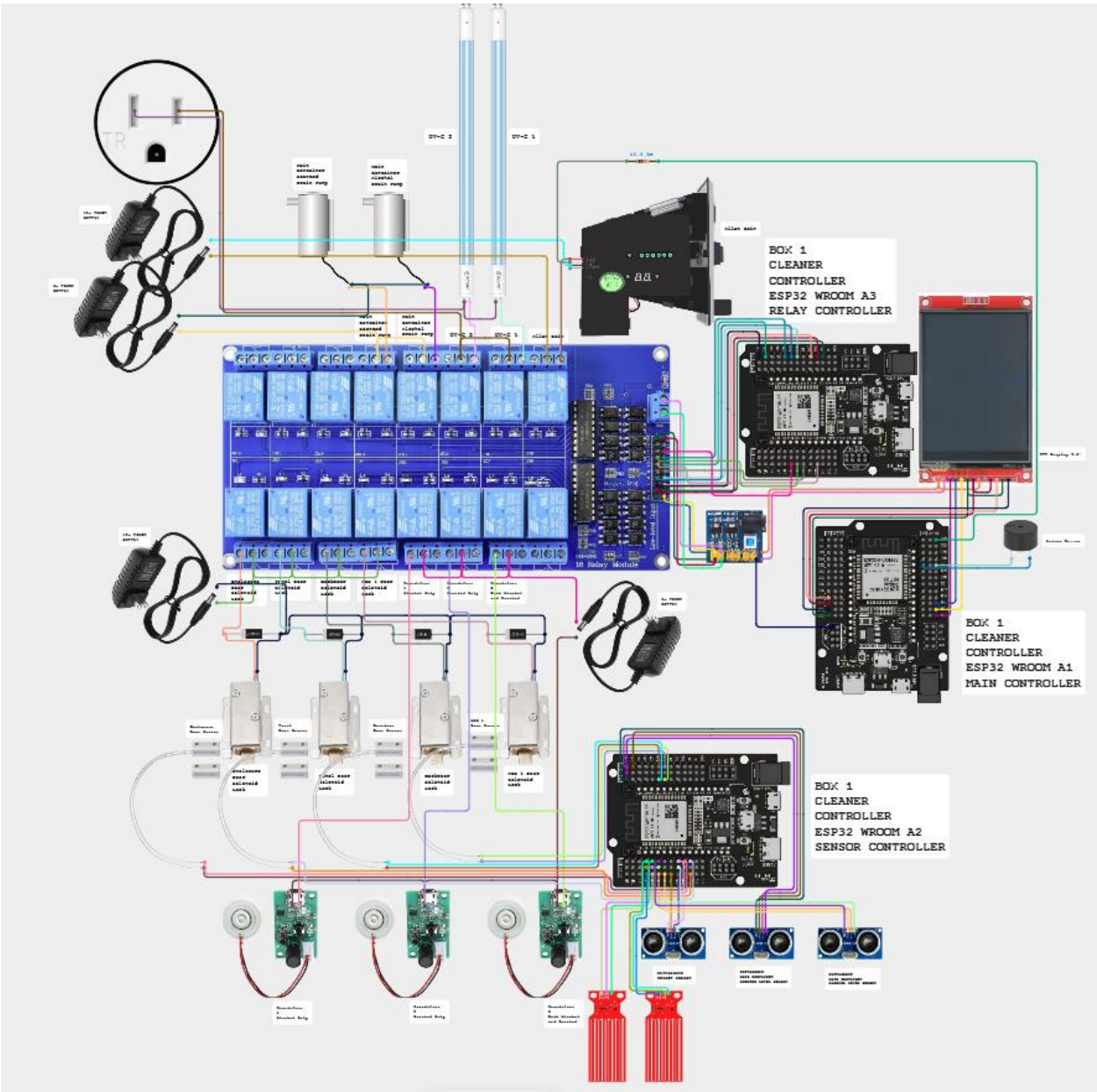
> The Cirkit drawing was checked against what was really built. Where the two differ, the pinout tables in the Technician guide are right (they come from the flashed firmware). Relay input numbers (IN1 ...) come from the drawing.

Whole installation



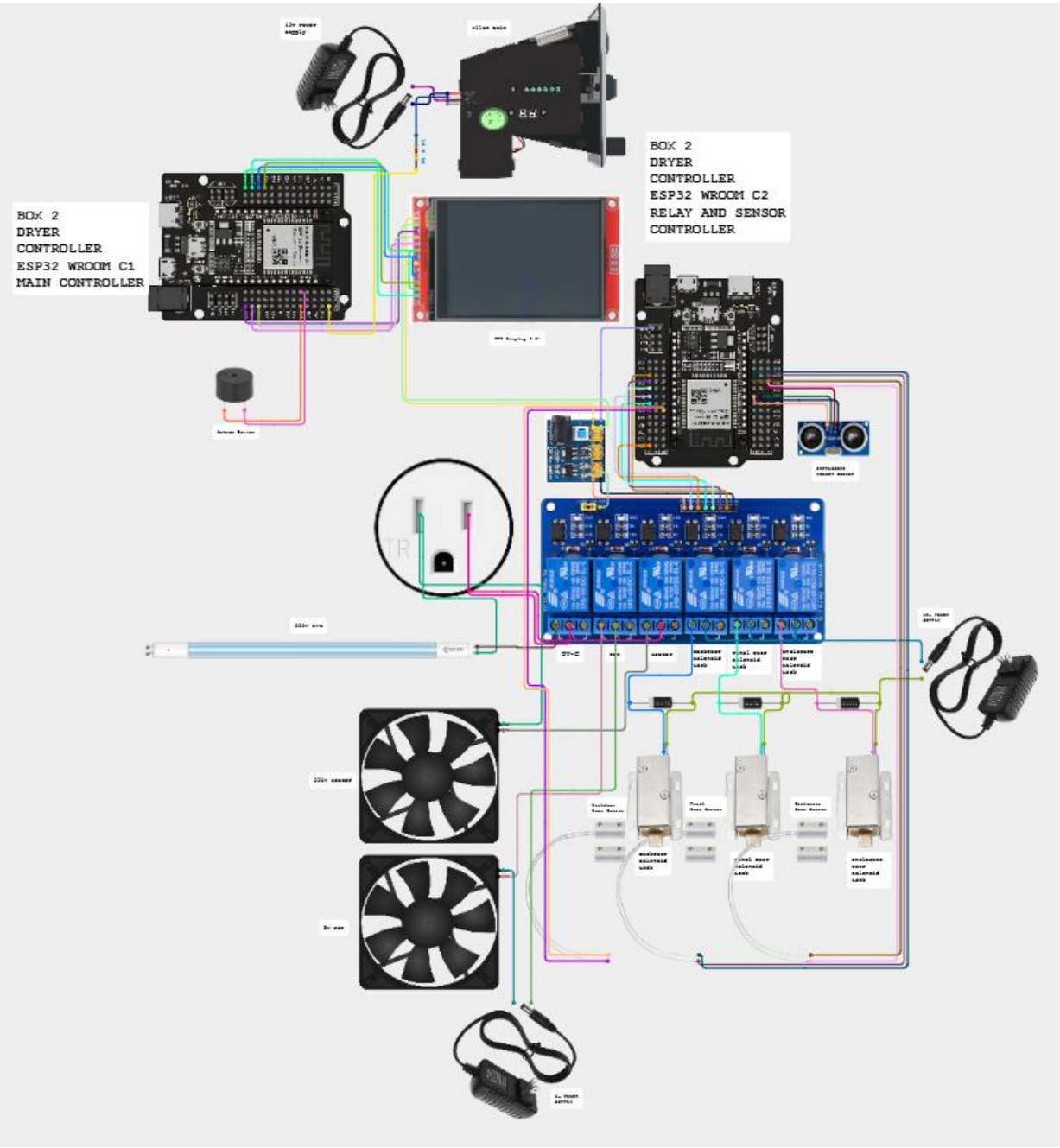
Whole installation (Cirkit Designer)

Box 1: A1, A2, A3



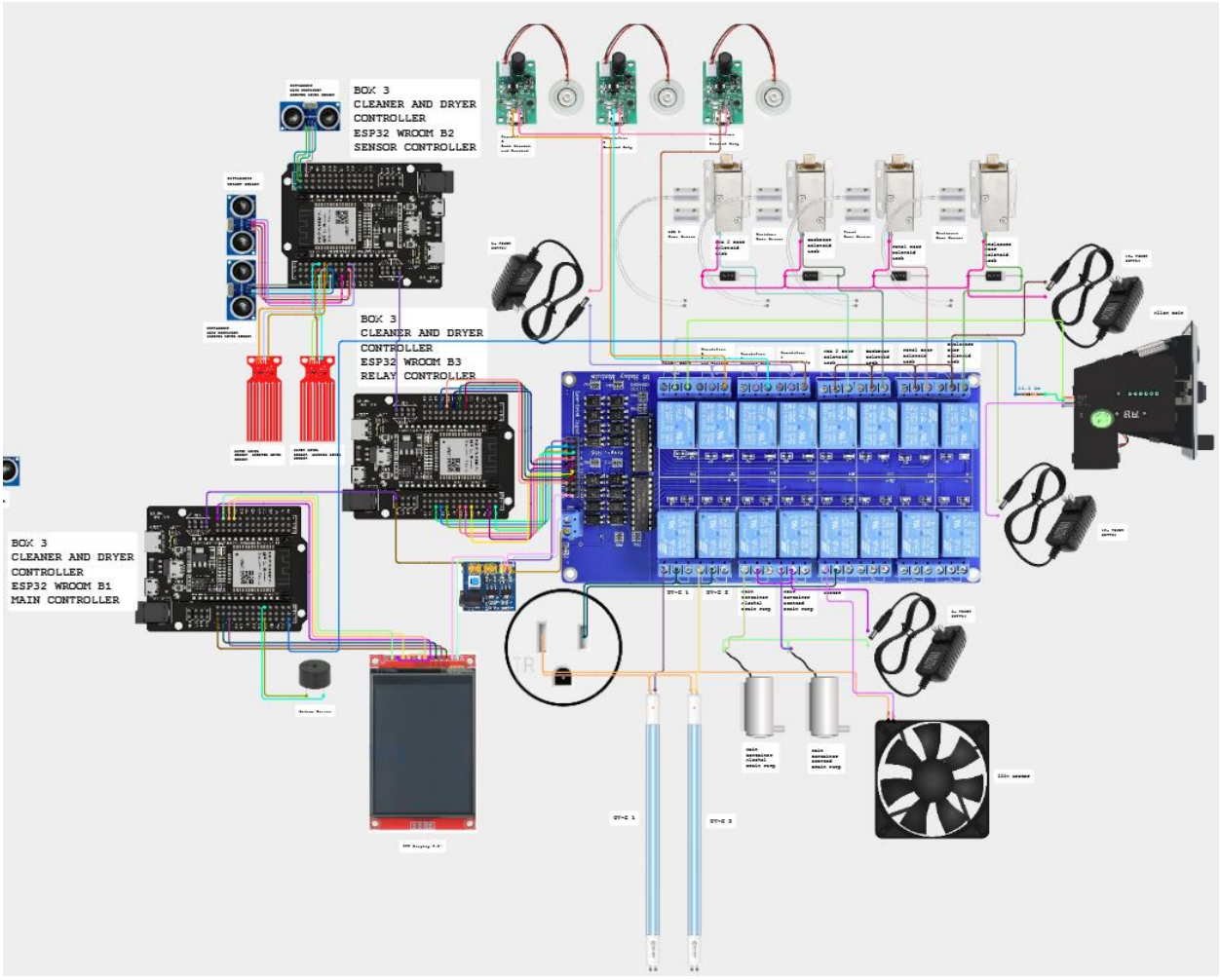
Box 1 (Cirkit Designer)

Box 2: C1, C2



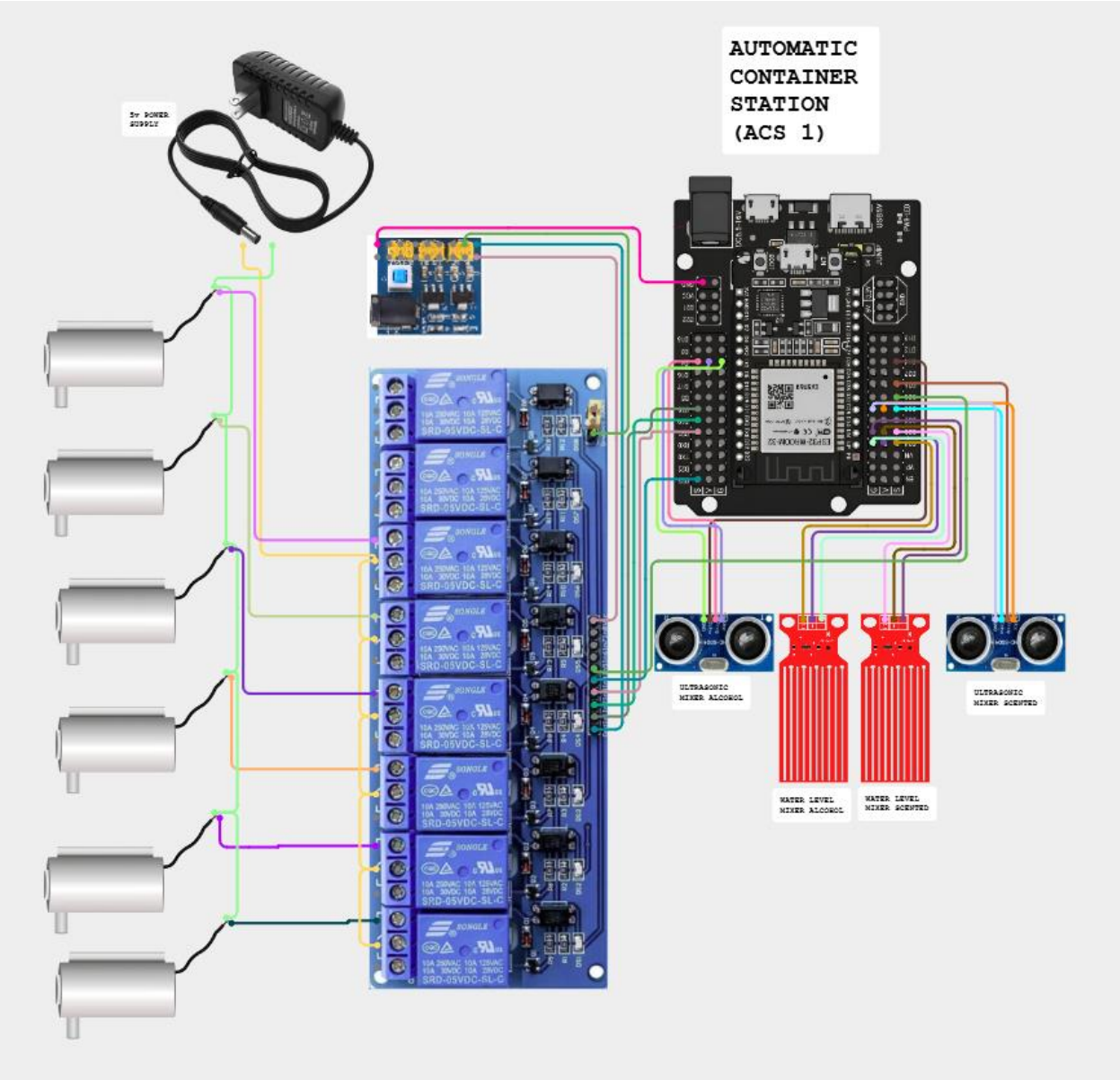
Box 2 (Cirkit Designer)

Box 3: B1, B2, B3



Box 3 (Cirkit Designer)

Refill station: ACS 1 and ACS 2



Refill station (Cirkit Designer)

Policies

What the machine does is fixed by the firmware. The rules around it are the owner's decisions. Items marked **confirm** are proposals: edit them to match what the owner decides, then post the customer rules at the machine.

Coins and refunds

Situation	What the machine does	Policy
Customer cancels a run	Asks CONFIRM or RESUME and warns that coins will not be refunded. The coins are **not** refunded.	No refund after a run starts.
Customer cancels while inserting coins	Asks YES or NO and warns that coins will not be refunded. The machine does not return coins already inserted.	No refund after inserting the coin
Fault or power cut during a run	The run stops, a trapped helmet is released, coins are not returned by the machine.	Staff offers a free run or a refund.
Machine OUT OF ORDER	It does not take coins.	A sign: use another machine.
Free retry	One free run after the final rating (default 2 minutes, for everyone). Not repeatable.	Set who gets it in Settings (lowest final rating).
Too few or too many coins	The time allot screen asks again. The most that can be bought is 90 coins in Box 1 and Box 3, 100 in Box 2.	Posted on the machine.
No change is given	The slot takes coins only.	Say so on the sign.

Using the machine

- Motorcycle helmets only. Take out gloves, phones, cameras and speakers first.
- Children use the machine with an adult.
- Never block, tape or remove the safety slide of the enclosure door. It is how a trapped helmet is freed when the lock or the power fails.
- The owner is not responsible for items left inside or for damage that comes from a helmet's own condition. Nothing is stored in the machine after a run.
- The machine cleans and dries. It does not guarantee that every smell or every drop of water is gone.
- Rating the helmet before and after is part of the run. The ratings are kept only as numbers.

Data and privacy

- The machines have no camera and no microphone. They record only counts: sessions, coins, minutes, ratings and the state of each board.

- No names, no phone numbers and no payment cards are stored. Payment is coins only.
- The Online Server sends only totals and status to Blynk. The Blynk token is a secret: keep it out of the manual, the web page and any file that is shared.
- The customer status page shows only whether each machine is free, in use or out of order. The public Blynk device behind it holds nothing else. The admin, technician and reference chapters of the manual are encrypted on the web and open with the team password.
- The Control Center and the setup Wi-Fi use a password printed in this manual. Change both before the machines are used in public, and give the admin and technician parts of this manual only to the team.

Staff and safety

- Only trained staff refill the containers, open the panels or use Maintenance.
- Mains work is for a qualified electrician (see Safety and emergencies).
- Run the weekly and monthly checks in the Maintenance chapter and write the date down.
- Alcohol and scented liquid: keep the supplier's safety data sheet near the ACS. Decide where the drain water may go.

Operating hours and notices

Item	Decision
Operating hours	9:00 AM – 5:00 PM
Where to report a fault or a lost item	MEJEZ STUDIO || 0975 791 5832
Price list posted at the machine	Prices and time chapter, or the printable panel cards.

Glossary

Words used in this manual, in plain language.

Term	Meaning
Acknowledge	The button on the online page that stops a repeating alarm. A new alert still sounds.
ACS	Auto Container System. The refill station: pumps and two mixer containers that top up a box's alcohol and scented containers.
Alert sound	The beep pattern of one kind of alert (siren, alarm, three beeps, two tones, chime...). Each can be changed in the Alerts tab.
Allan coin acceptor	The coin slot. It sends one pulse for each coin value it accepts.
Blynk	The online service the Online Server sends its snapshot to. The web page reads it from there.
Brain	The ESP32-S3 board of a box (S3A, S3B, S3C). It runs every step and keeps the settings. A box has one brain and several terminals.
Captive portal	The page that opens by itself when a phone joins the setup Wi-Fi of the Online Server.
Channel 1	The radio channel all the boards use for ESP-NOW. The LocalServer Wi-Fi stays on it too.
Control Center	The pages of the LocalServer (Home, Box 1, Box 2, Box 3, ACS) where settings, calibration and Maintenance are done.
Cool down	The time after heating when nothing heats and the helmet can cool. Box 2 runs the fan; Box 3 lets everything rest.
Customer status page	status.html: the public page that shows which machine is free, in use or out of order.
Datastream	A named value in Blynk. This project uses V0 (a text) and, if you switch them on, V1 to V4.
DC-DC module	The power module that turns 12 V into 5 V and 3.3 V for the boards, sensors and small loads.
Early warning	The level (30 % by default) where a container dial turns amber and a notice appears, before the box runs out.
Enclosure lock	The solenoid lock of the main door of a box. It is the one released for a trapped helmet.
ESP-NOW	The radio link the boards talk over. No router is needed.
ESP32 and ESP32-S3	The two kinds of board. The plain ESP32 boards are terminals. The S3 boards are brains and servers.
Free retry	One free run after the final rating (2 minutes by default).
GPIO	A numbered pin of an ESP32 board.
Humidifier	The ultrasonic mist maker that makes the alcohol and scented mist (5 V).
Hysteresis	The gap between the WET and the DRY reading of a water sensor, so it does not flicker.

Term	Meaning
JD-VCC	The separate power input of a relay module's coils, with a jumper to the logic supply.
Level dial	The half-circle gauge of a container on the Monitor tab. 0 is the refill line.
LocalServer	The board that makes the ESP32_LOCAL Wi-Fi and the Control Center, and counts the revenue.
Maintenance mode	A page where a technician switches relays or pumps by hand. The machine shows MAINTENANCE.
Mixer	One of the two containers of an ACS (alcohol, scented) where the recipe is made before it goes to a box.
Mist	The alcohol or scented vapour used in cleaning. Mist method: balanced, more alcohol, more scented.
NVS	The memory inside a board where settings are saved (they survive a power cut).
Online Server	The ESP32-S3 that listens to the machines and sends a snapshot to Blynk. It never controls anything.
OUT OF ORDER	The screen shown when a box cannot serve customers.
Phase	A part of a cleaning run: start, middle or end. Each relay is on or off per part.
Power-to-unlock	A lock that opens only while it has power. With no power the door stays locked.
Public status device	The second Blynk device whose token may be public. It holds only the step and time left of each box.
QR code	The square code on the sign at each machine. A phone camera opens the customer status page with it.
Rating	The 1 to 10 score the customer gives the helmet before and after: odor (Box 1 and cleaning in Box 3) or dryness (Box 2 and drying).
Relay	An electrically operated switch. The boards use relays to switch locks, pumps, mist makers, lamps and the heater.
SAFETY PAUSE	The screen shown when a door opens during a run. The loads switch off until the door is closed.
Safety slide	The mechanical slide at the lower left of the enclosure door. Slide it down and the door unlocks by hand, with no power and no help from the boards.
Serial Monitor	The Arduino IDE window that shows what a board prints over USB (115200 baud).
SoftAP	A Wi-Fi network made by a board itself (ESP32_LOCAL, JollyHelmet-Online).
Team password	The password that opens the encrypted admin, technician and reference chapters of the Manual tab.
Terminal	A plain ESP32 board that holds a box's sensors, relays and display. It follows its brain.
TFT	The 2.8 inch colour touch screen of a box.
Time allot	The screen where the customer types the minutes to buy.
Token	The secret key that lets the Online Server write to your Blynk template. Keep it private.

Term	Meaning
Ultrasonic sensor	The HC-SR04. It measures the distance to the helmet or to the liquid in a container.
UV-C lamp	The 220 V germicidal lamp. Never look at it when it is lit.
Watchdog	A safety check: if a board stops hearing its brain for 15 seconds it switches its loads off.

Version and revisions

Item	Value
Manual	Jolly Helmet User Manual, revision 7, 2026-10-11
Made by	MEJEZ STUDIO
Covers	Box 1, Box 2, Box 3, ACS 1, ACS 2, LocalServer, Online Server
Protocol file	Shared_Common.h, identical in every sketch folder (checksum b9da9bfb)

Revision history

Rev	Date	What changed
1	2026-10-07	First manual: customer guide and admin guide for Box 1, 2 and 3, prices, Control Center, ACS. PDF and Word, new logo, Arial.
2	2026-10-07	Technician guide: components, pinouts, circuit and wiring, firmware and flashing, maintenance, hardware troubleshooting. Corrected from the Cirkit Designer document to what was really built.
3	2026-10-10	Online Server, Blynk setup and the web page. Real parts: UV-C 220 V lamps, HC-SR04, 5 V mist makers and pumps, DC-DC module. Team names on the cover.
4	2026-10-10	Reference section: safety and emergencies, box flowcharts, troubleshooting decision trees, settings reference, schematic pages, policies, glossary, this page and the index.
5	2026-10-11	Online page: level dials with early warning, Alerts with a sound for every kind of alert, Reports (busiest hours and days, what each box earns), Checklist, Fix it, customer status page and QR sign, Manual tab with a team password. New Online Server firmware (per-box statistics, public status device).
6	2026-10-11	The safety slide of the enclosure door (lower left, slide down to unlock by hand when the solenoid lock is broken or there is no power): customer guide, admin guide, technician guide, safety and emergencies, decision tree 4, settings reference, glossary and policies.
7	2026-10-11	Box flowcharts and troubleshooting decision trees redrawn in black and white (no colours): thick border = the machine is running or "do this", double border = a fault, black box = start or end.

Firmware in this manual

The date is when the sketch file was last saved. The board in the machine may carry an older build: compare with what it prints when it starts (Serial Monitor, 115200).

Sketch folder	Board	Role	File date
ESP32_S3A	S3A	Box 1 brain	2026-10-05
ESP32_A1	A1	Box 1 screen, coin, buzzer	2026-10-02
ESP32_A2	A2	Box 1 sensors	2026-10-02

Sketch folder	Board	Role	File date
ESP32_A3	A3	Box 1 relays	2026-10-02
ESP32_S3C	S3C	Box 2 brain	2026-10-05
ESP32_C1	C1	Box 2 screen, coin, buzzer	2026-10-03
ESP32_C2	C2	Box 2 sensors and relays	2026-10-03
ESP32_S3B	S3B	Box 3 brain	2026-10-05
ESP32_B1	B1	Box 3 screen, coin, buzzer	2026-10-03
ESP32_B2	B2	Box 3 sensors	2026-10-03
ESP32_B3	B3	Box 3 relays	2026-10-03
ESP32_ACS1	ACS 1	Box 1 refill station	2026-10-03
ESP32_ACS2	ACS 2	Box 3 refill station	2026-10-03
ESP32_LOCALSRV	LocalServer	Control Center, revenue	2026-10-09
ESP32_ONLINESRV	Online Server	Blynk monitor	2026-10-10

Tools and libraries

Item	Version
Arduino IDE board package: esp32 by Espressif	3.3.7
TFT_eSPI	2.5.43
ArduinoJson	7.4.3
ESP_Async_WebServer	3.12.0
Async_TCP	3.5.0
Blynk	Blynk Cloud HTTPS API (no library on the board)

Check before you hand over

- Measure the real containers, enter the LEVELS distances and the ACS lines, then switch **Check levels** and **Ask the ACS to refill** on.
- Run every relay and every pump by hand in Maintenance before the first automatic run.
- Calibrate the touch screen of B1 (it still has the A1 numbers) with ESP32_C1_CALIBRATE.
- Online Server: after its first upload, the Control Center must still hear every board (13 of 13). If it does not, check the radio channel.
- Online page: flash the Online Server firmware of 2026-10-10, create the public status Blynk device, print the QR sign for every machine, and run build_web_manual.ps1 after every change to this manual.

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Thank you for using
Jolly Helmet

We hope to see you again!
Make your Helmet Happy Again.

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DEVELOPERS AND DESIGNERS

USER MANUAL